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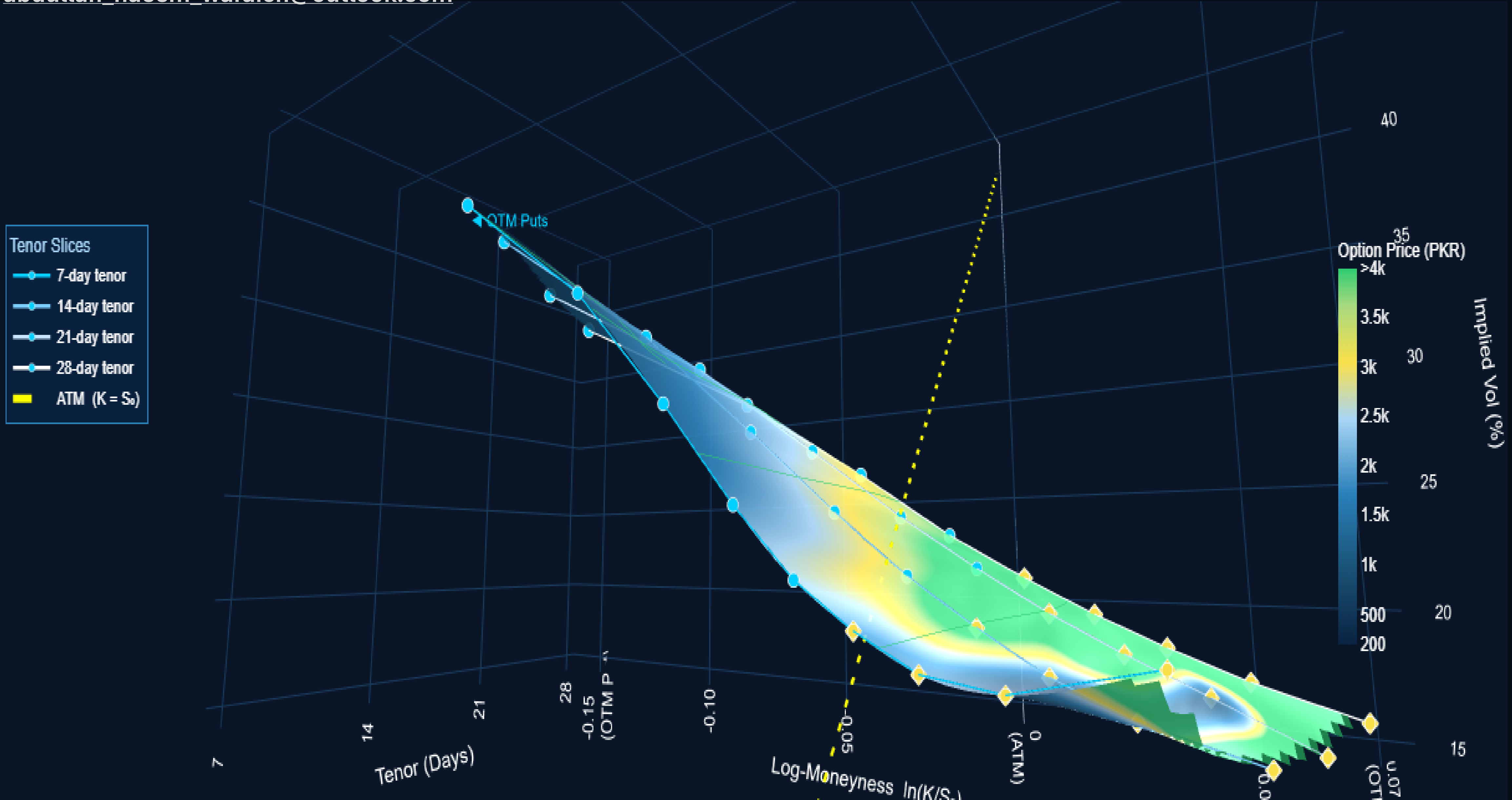
VOLATILITY TRADING FOR PAKISTAN

KSE-100 Index — Complete Volatility Surface (Log-Moneyness)

Puts (blue ●) + Calls (yellow ◆) | Arbitrage-Free | 8-May-2026 | S_0 = 171,704 PKR



ACKNOWLEDGEMENT

This work was completed during a three-month engagement as an Options Risk Intern at the **National Clearing Company of Pakistan (NCCPL)**.

For nearly a decade, **PSX** and **NCCPL** have pursued the launch of exchange-traded options in Pakistan — a milestone requiring two problems to be solved simultaneously: arbitrage-free pricing at inception with correct volatility skew and smile, and a robust margining framework for risky volatility strategies. Both were solved within this engagement.

The pricing framework stands on a hard lessons: the **traders who lost everything in the 1987 crash** — teaching the world that skew is not optional, it is survival — and a foresight by **Bruno Dupire**, whose break-even implied volatility framework gave us the mathematical machinery to build arbitrage-free smiles from day one, without a single traded option to calibrate against.

The margining framework gives NCCPL the risk infrastructure to safely clear strategies Pakistan's markets have never seen before.

My gratitude to Chief Risk Officer **Kashif Alam** for the intellectual trust, and to **Ali Hassan** for the data.

The options market in Pakistan will launch one day. When it does, it will be priced correctly — and the smiles will be real.

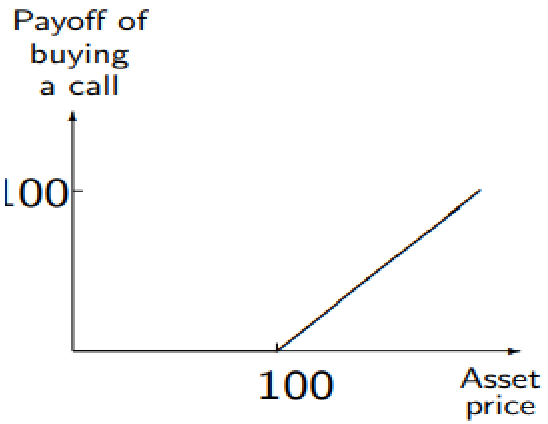
CONTENT

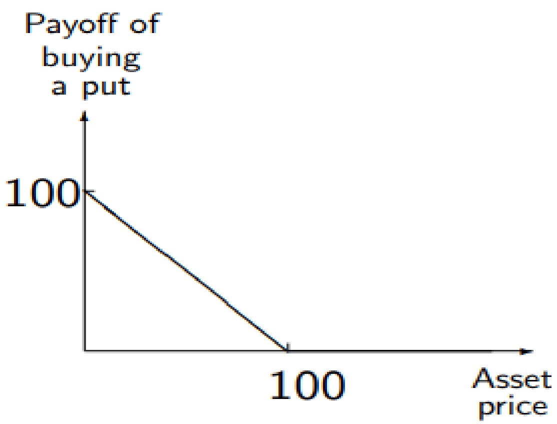
Options: Volatility Instrument

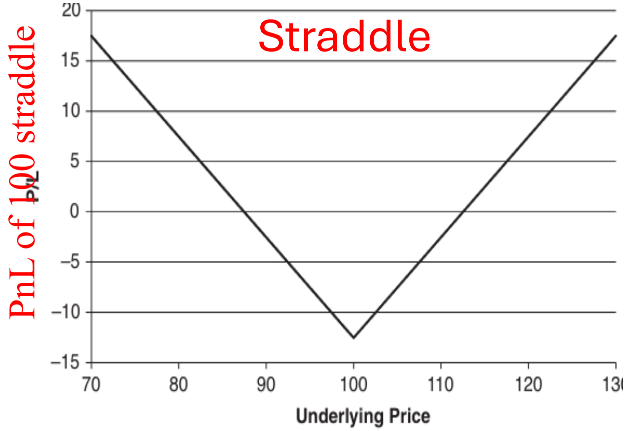
- Call(Put): Right (but not obligation) to purchase(sell) a given asset at a future date – called expiry/maturity for a price agreed today –called strike. Call holder will exercise at maturity if underlying price is greater than strike: payoff = S-K. Put holder will exercise at maturity if underlying price is smaller than strike: payoff = K-S. Via **put-call parity**, we can derive no-arbitrage price of a put from call option with same strike and time to maturity: so we will focus on call in these notes.

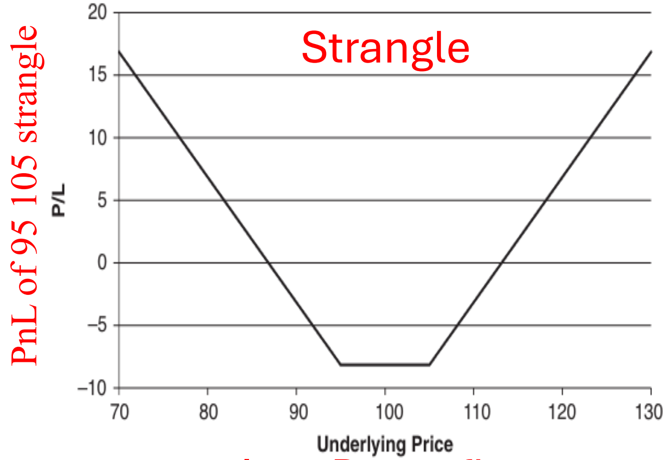
- Call price depends on spot price (distance to strike and maturity) Option price must increase with maturity. Hence, we expect call value to decrease with time(keeping all other variables fixed). Call convex in spot.

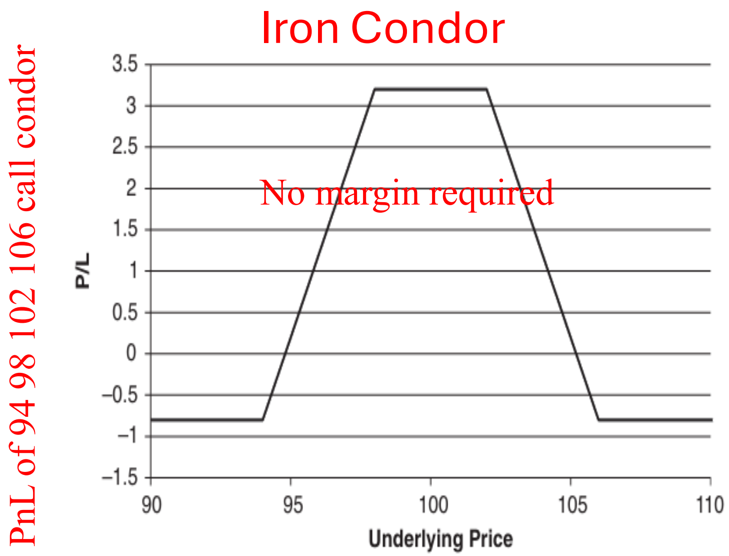
- *From directional bets to bets on volatility*:
  - **Long Volatility**: Independent of stock rallying or plummeting by buying a straddle(ATM call and put combination i.e., strike prices equal initial spot price of underlying), one can reap enormous profits under market crashes and extremely volatile periods. **Index plummeted** in the month of Feb 2026**;** **long straddle** strategy resulted in a **profit** of **Rs. 2100000.** **Long strangle**: buying an OTM put and call, with put strike less than call's.
  - **Short Volatility:** Short 100 straddle and long 95 105 strangle giving us a 95 100 105 **butterfly:** profitable when stock stays close to where it started. Long 94 & 106 call , while short 98 & 102 call gives us **Call Condor**

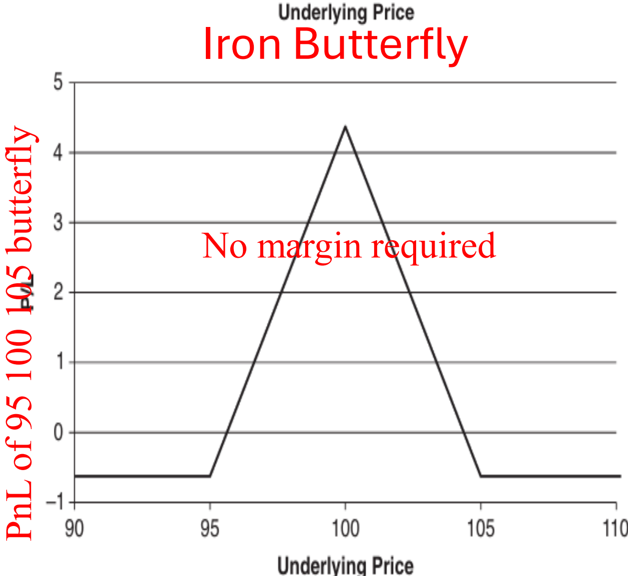










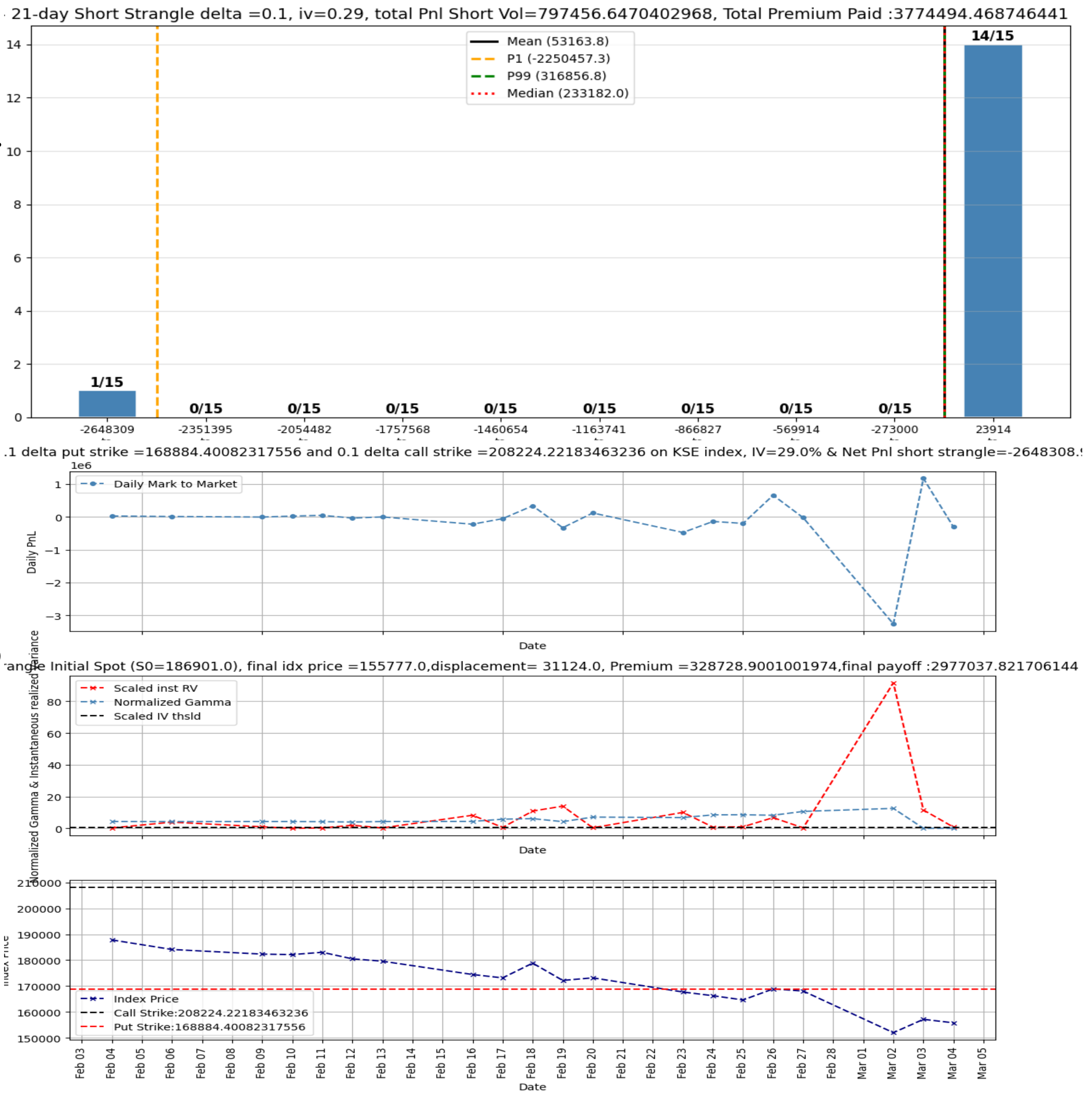


Liquidity Under Market Stress/ Risk Appetite

- Stock market in a downtrend since Feb 2026: how to provide liquidity when stock market is in a downward spiral? Majority of retail traders are risk-averse, losses hurt them more than profits and that is why they buy insurances. Deep out of the money puts offer such an insurance.

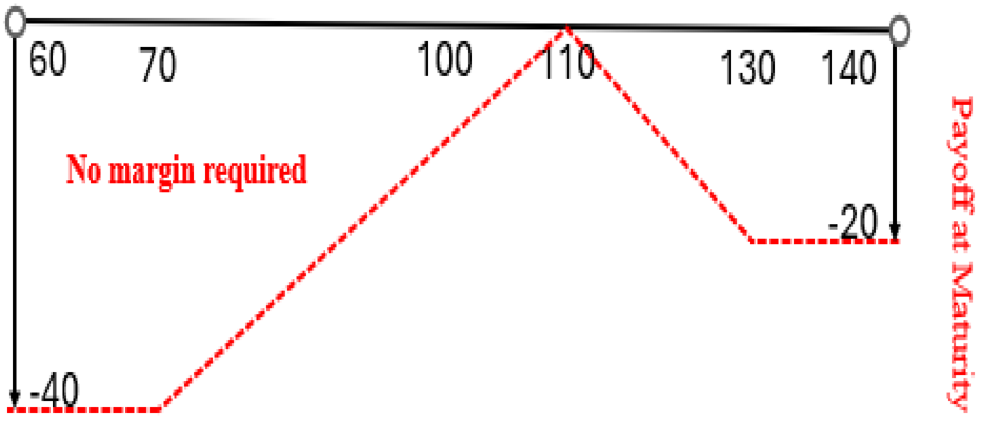
- People actively buy health, car and life insurances paying a premium over fair value as humans tend to over-estimate the probability of adverse events happening. Consequently, there numerous monthly insurance premiums feel like nothing when once in a lifetime their insurance pays out in large sums: even **though in expectation they may have a net loss**.

- **Tail Risk Protection:** This phenomenon can be realized from insurance seller short volatility position(selling a deep OTM call and put with life cycle of 1 month and IV=29%) for last 15 months on KSE-100 index. 14/15 times insurance seller made money as option was not exercised 14/15 times. But in time of crisis when index plummeted in Feb 2026 insurance buyer had a net profit of Rs. 2.6 Million. From utility POV Rs. 2.6 Million in crash worth much more than same amount in a boon. Therefore, even with a net loss of Rs 0.8 Million retail trader won't complain.

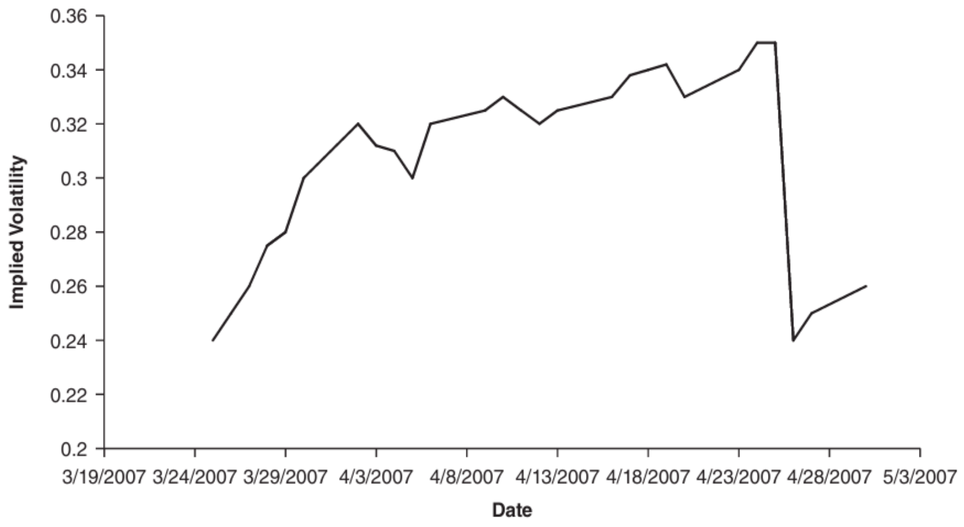


Risk Appetites & Implied Volatility

- Short straddle is extremely risk; to bound one's downside without compromising on short volatility opinion we can go **Long Iron Butterfly with Asymmetric Wings** which includes shorting an ATM straddle 110, buying a 70 Put and 130 Call.
- A **Call Condor** is a position where we buy a call, sell a call with a higher strike, sell a call with a still higher strike, and buy another call with an even higher strike (we can also construct a put condor in an analogous fashion). The strikes should be equally spaced.
- **Intrinsic value**: immediate worth to the long side of the contract? For a call option: moneyness of an option is proportional to positivity of $\log\left(\frac{\text{Spot Price}}{\text{Strike Price}}\right)$, if spot is smaller than strike option has no intrinsic value
- **Extrinsic value**: expected future option worth. More time to expiry, more extrinsic value of the option. For option C^1 and C^2 on the same underlying, with same strike, with time to maturity for C^1 is 15 days and time to maturity for C^2 is 30 days, although both have same intrinsic value but C^2 should have a higher price because of higher extrinsic value.
- Extrinsic value of an option is mostly influenced by the **<u>Implied Volatility</u>:** future expected volatility of the underlying during the life-cycle of the option. <u>An option on AAPL expiring immediately after earning release has much higher risk for seller compared to one expiring before earning, so premium he charges already accounts for this.</u>



Final Payoff for Asymmetric Long Iron Butterfly: short straddle 110, long 70 put and 130 call

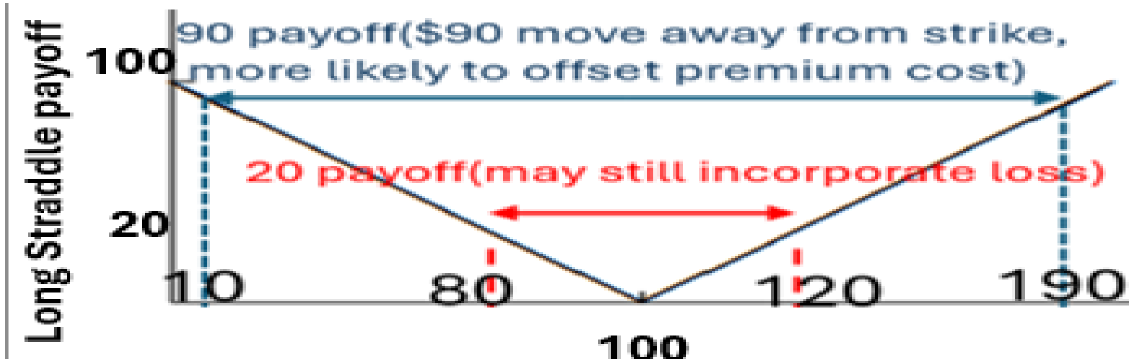


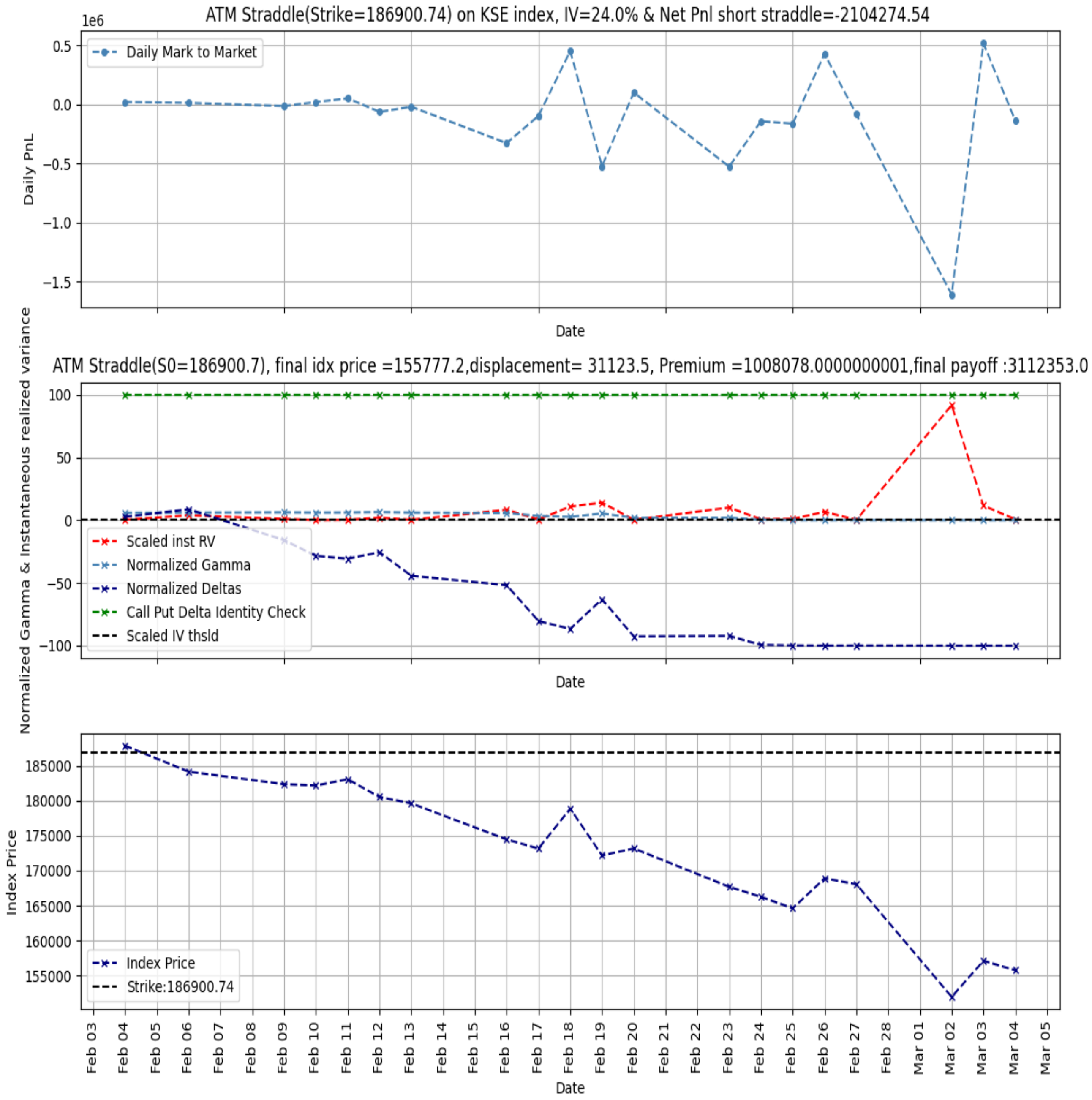
The AAPL Front-Month Volatility around the Second Quarter 2007 Earning Release

Volatility Trading: Short Straddle PnL

- Convexity of options price w.r.t strike. Call(Put) has a positive payoff at maturity if stock price > strike :=100 (stock price < 100)
- Long ATM straddle makes money when absolute difference between final stock price S_T and strike K = 100 is greater than premium he paid for call & put(assume interest rate=0): options can be combined as to have a **long/short volatility position**. S_T = 95,105 same straddle payoff
- Option seller's view of future expected volatility(IV) is priced inside premium, if stock realizes large moves away from strike i.e., realized volatility > implied volatility, buyer Of straddle makes money
- Unhedged Short Straddle and Short Put are lottery tickets for long vol positions. KSE100 index price during the month of Feb 2026 dropped from 186000 → 150000: a short straddle position would have incurred **Rs. 2 Million loss**

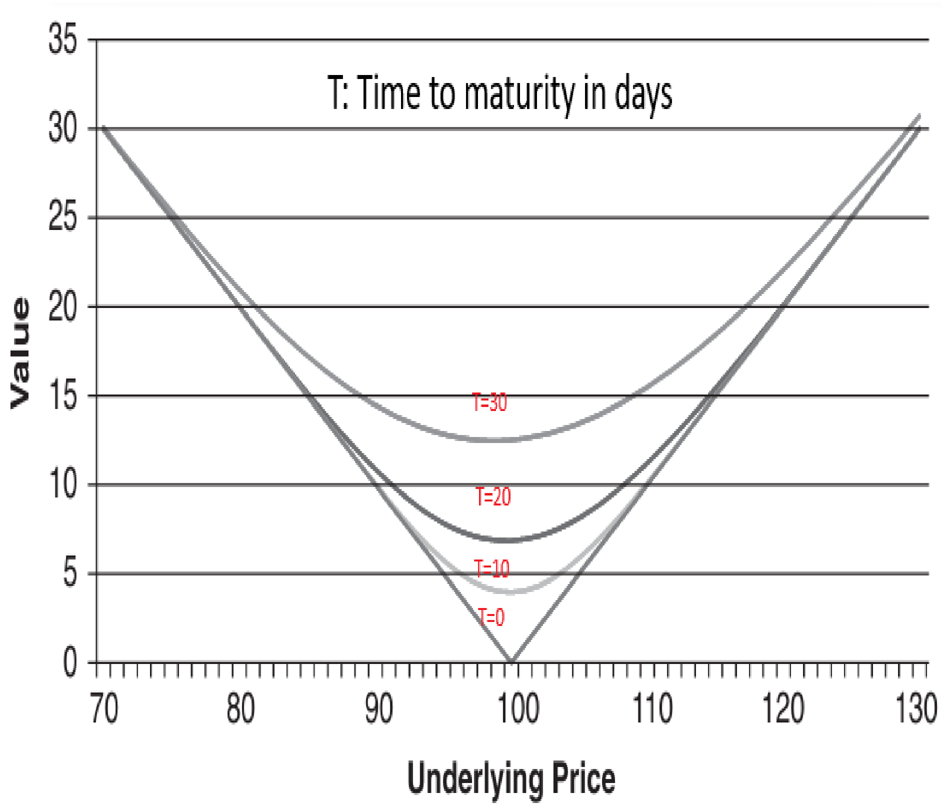
Options are zero sum game, buyer's profit=seller's loss





Vega Risk & Theta Bleed

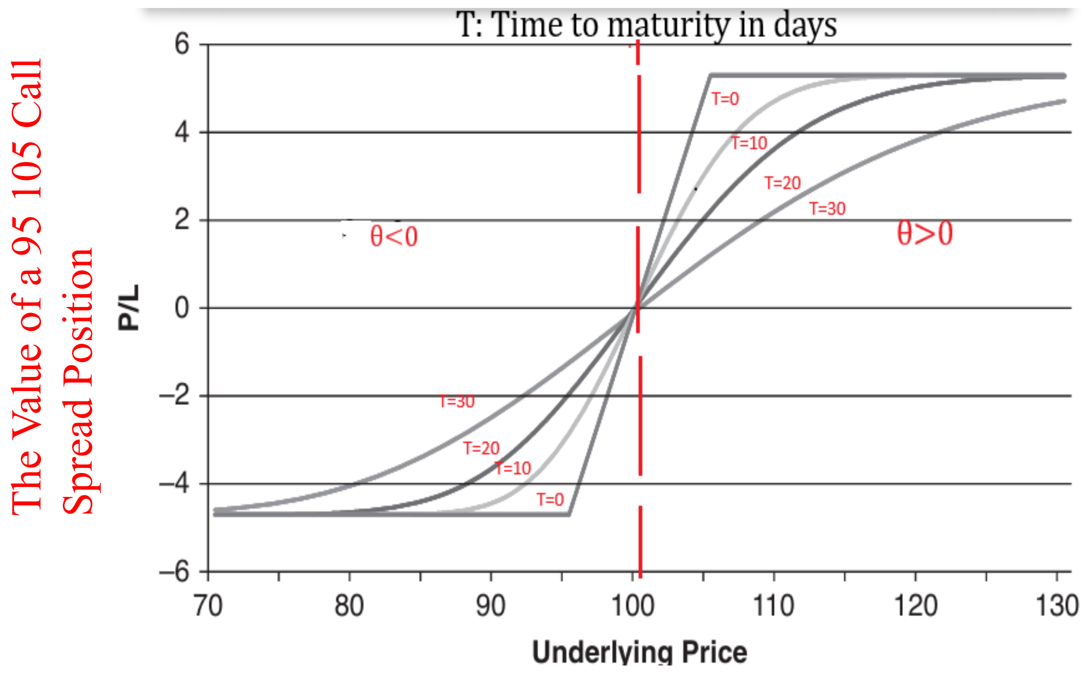
- Long option position corresponds to negative θ i.e., option loses its' extrinsic value as it approaches maturity because probability of an extreme move happening in 7 days is less compared to 1 year; often called as theta bleed for long option.
- Option price is convex in Strike. These observations can be verified from 30 days **Straddle** price dynamics shown on the right.
- Recall that a naked option seller is prone to unbounded losses, whereas the buyer at worst loses premium, so it is only fair to him if he charges a premium proportional to his expectation of uncertainty over the lifetime of option(finance jargon **implied volatility** which is one of the inputs to BSM model**)**.
- **Vega:** <u>Sensitivity of call/put price to changes in implied volatility is called Vega.</u> <u>An option seller shorts an option in the morning, few minutes later a global breaking news(US-Iran war) increases panic in the market; consequently, option price increases and long position is worth more now (short seller's liability increases) i.e., option buyer is</u> **Long Vega** and seller **is Short Vega**.
- Short volatility position is extremely prone to negative market sentiments. **Vega risk** is more of a concern for longer dated options

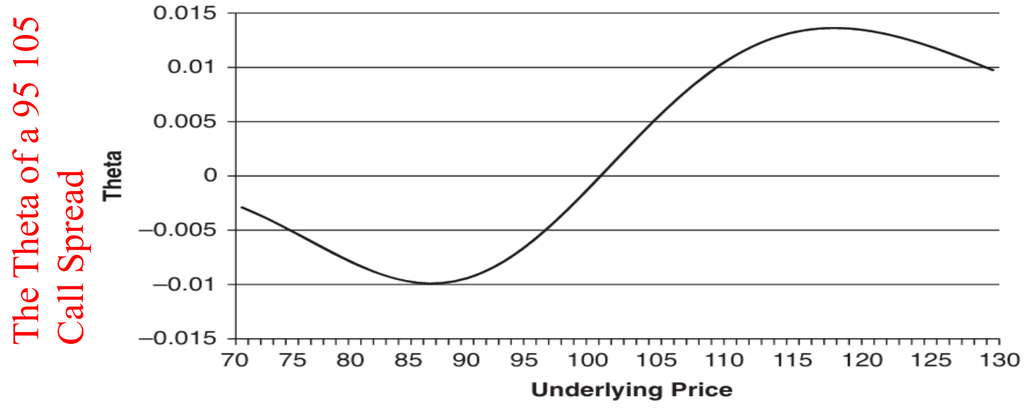


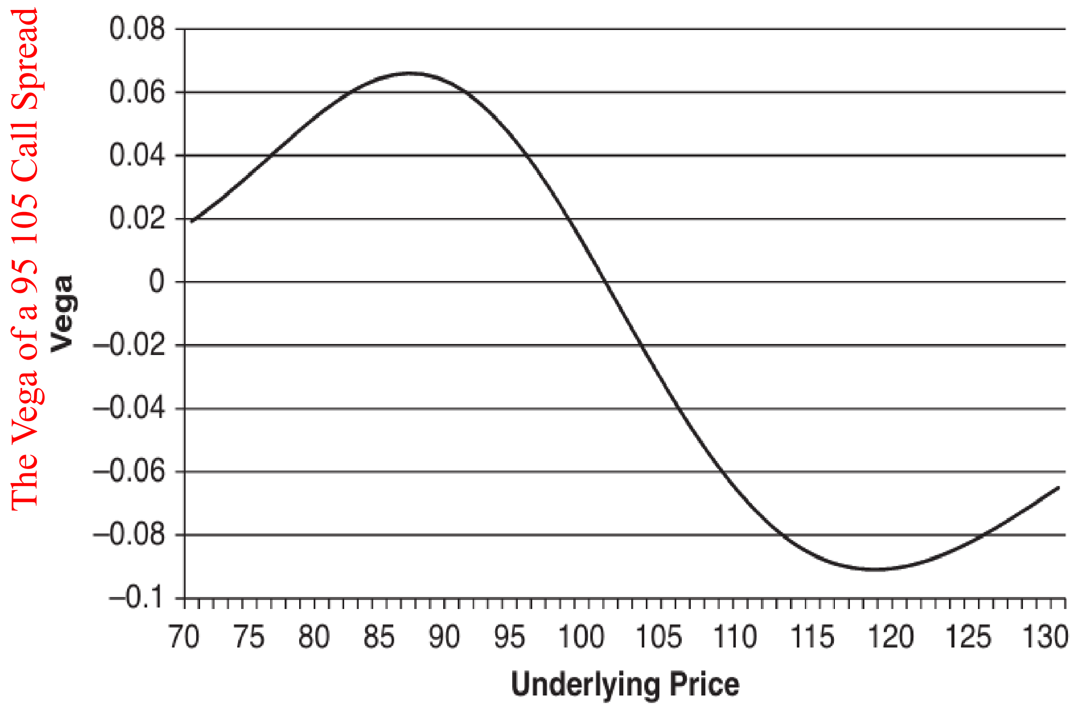
The Dependence of the Straddle Value on Time

Theta & Vega Dependence On Spot

- **Call Spread:** A long call spread when we buy a call(strike=X_1) and sell another with a higher strike X_2. If spot is near X_2, the short position we should have $\theta > 0$. Near X_1, we have negative θ . Similarly, spot around lower strike is long Vega(higher strike short Vega). Max [loss](prof) equals $[C(X_1) - C(X_2)](X_2 - X_1 - C(X_1) + C(X_2))$.
- If the spot is at 80(deep OTM for both calls), an increase in implied volatility implies larger jumps in stock price, increased probability of spot being pushed ITM w.r.t lower strike X_1. Therefore, for $x \leq 95$, we see that Vega: $= \frac{\Delta C}{\Delta \sigma_{iv}} \geq 0$. On the other hand, if spot is in the range $x \geq 105$, call spread has a maximum pay-off and larger moves(higher volatility) increases the probability of pushing spot into smaller pay-off region $\theta < 0$ implying negative Vega. When $x \gg 105$, both strikes are deep ITM → near zero Vega







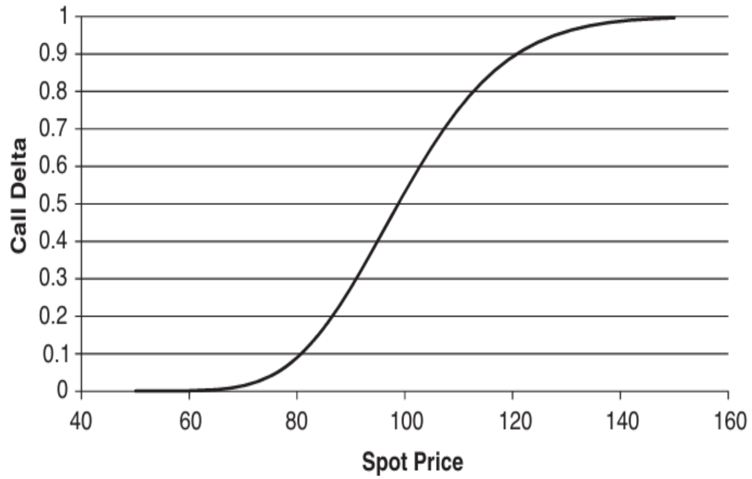
Directional Risk

- **Delta *Δ*** is sensitivity of option price to changes in underlying's price. $\Delta = 0.40$ means that a 100 USD increase in stock price leads to an increase of 40 USD in options price. can be interpreted as 40 % chance of option expiring ITM. For call(put) option, delta is an increasing(decreasing) function of spot price

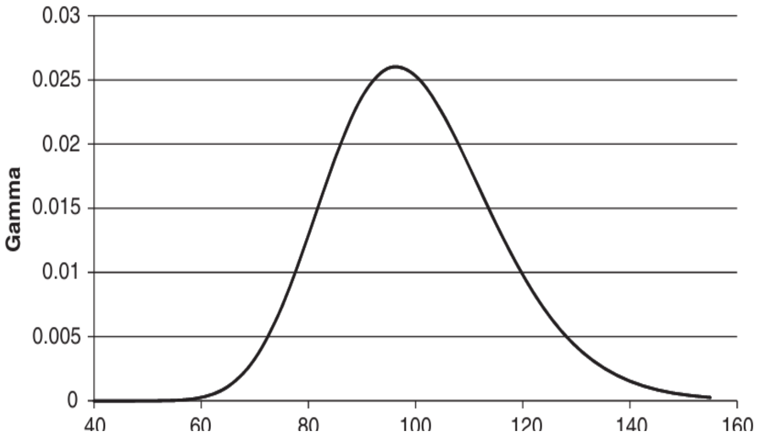
- **Gamma Γ**, measures sensitivity of option prices to changes in delta. Example: Call Premium = \$2.00, Spot = \$13.00, Delta = 0.40 and Gamma = 0.05. If spot moves from \$13→ \$14, then call price moves from \$2.00→ \$2.40 and Delta moves from 0.40→0.45($\frac{\text{New Delta}-\text{Old Delta}}{\text{New Spot}-\text{Old Spot}}$ =Gamma). Now spot moves again \$14→ \$15 implying call price moves from \$2.40 → \$2.85.

- If option seller has not hedged against the directional moves in the underlying he is exposed to **directional risk**: an expected drop in price tomorrow will result in loss scaled by :
$\Delta(\text{put}) \times \text{Notional} \times (\#\text{ sold contracts}) \times (\text{New Spot} - \text{Old Spot})$

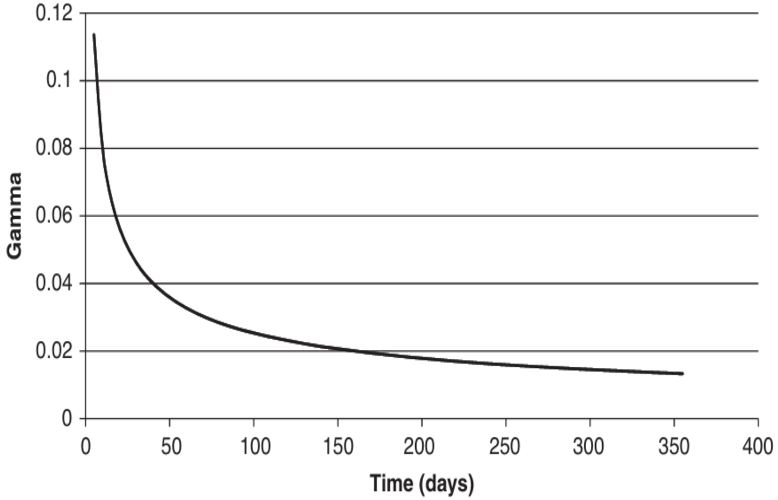
Delta of a call option with strike 100, time to maturity 1 year and implied volatility of 30%



Gamma as a function of Spot Price, time to maturity is 1 year and implied volatility of 30%



Gamma as a function of time to maturity for an ATM option.



Black Scholes Merton(BSM) Pricing Model For Risk Analysis

- We don't need a model to speculate on volatility. A straddle is a volatility bet. Models are essentially an arbitrage-free way to convert the quickly changing option prices into a slowly changing parameter

- We would like to find out how value of the long option portfolio varies as time, stock price and implied volatility changes from $t \rightarrow t + dt$, $S_t \rightarrow S_{t+dt}$ and $\sigma_{IV}^{t} \rightarrow \sigma_{IV}^{t+dt}$ respectively. We can use Taylor's approximation to get:

$$C\left(t + dt, S_{t+dt}, \sigma_{IV}^{t+dt}\right) - C(t, S_t, \sigma_{IV}^{t}) = \theta_t dt + \Delta_t(S_{t+dt} - S_t) + \frac{\Gamma_t S_t^2}{2}\left(\frac{S_{t+dt} - S_t}{S_t}\right)^2 + \mathrm{Vega}_t\,(\sigma_{IV}^{t+dt} - \sigma_{IV}^{t})$$

- Black Scholes Merton give us an explicit formulas for the Greeks: $\theta_t, \Delta_t, \Gamma_t$ *and* Vega_t as a function of time, underlying stock price and implied volatility when combined with historical estimates on inter-day return volatility squared $\left(\frac{S_{t+dt} - S_t}{S_t}\right)^2$ and changes in implied volatility $(\sigma_{IV}^{t+dt} - \sigma_{IV}^{t})$, provide us Value At Risk for options portfolio. For possible losses over a single day horizon $dt = \frac{1}{252}$. We will assume implied volatility does not change over the lifetime of the option.

- Market forces decide option prices, every option contract has explicit listing of maturity, strike and spot price is available from exchange, which we will denote by C^{MKT}. **Option's Implied volatility:** σ_{IV} is a volatility parameter when plugged into Black Scholes Model satisfies: $C^{BS}(K, S, T, r, \sigma_{IV}) \approx C^{MKT}$.

Gamma Risk & Variance Premium

- For strike K, interest rate $r = 0$, current stock price S_0 and assuming 252 business days; option seller collects $C_0 = \text{Premium} = BS(S_0, K, T = \frac{20}{252}, r = 0, \sigma_{IV})$, where σ_{IV} is the options implied volatility, for writing a call option expiring in 21 business days from now. He buys Δ_0^{BS} shares of the underlying stock, so he is not exposed to any directional changes. His Mark to Market on the day $t + dt \leq \frac{20}{252}$, $MtM(t + dt) = -\theta_t dt + \frac{\Gamma_t S_t^2}{2}\left(\frac{S_{t+dt} - S_t}{S_t}\right)^2$, $-\theta_t$ is always positive works in favor of short option position and for long option it is called **theta bleed**.

- According to Black Scholes Merton partial differential equation(interest compounding negligible over less than a month), option loses a value of $-\theta_t dt$ as time to maturity decreases from $(T - t) \rightarrow (T - t - dt)$ and this loss in value satisfy $-\theta_t dt = \frac{\Gamma_t S_t^2 \sigma_{IV}^2}{2}$. Plugging it back into mark to market:

$$MtM(t + dt) = -\theta_t dt + \frac{\Gamma_t S_t^2}{2}\left(\frac{S_{t+dt} - S_t}{S_t}\right)^2 = \frac{\Gamma_t S_t^2}{2}\left(\sigma_{iv}^2 dt - \left(\frac{S_{t+dt} - S_t}{S_t}\right)^2\right),$$

That is to say tomorrow's PnL for any option position is proportional to Gamma.

- **Variance Premium:** also known as the volatility premium is the tendency for implied volatility to be higher than subsequently realized volatility. As narrated by Euan Sinclair: In his 1906 book The Put and Call, Leonard Higgins writes how traders on the London Stock Exchange first determine a statistical fair value for options, then "add to the 'average value' of the put and call an amount which will give a fair margin of profit." That is, a variance premium was added.

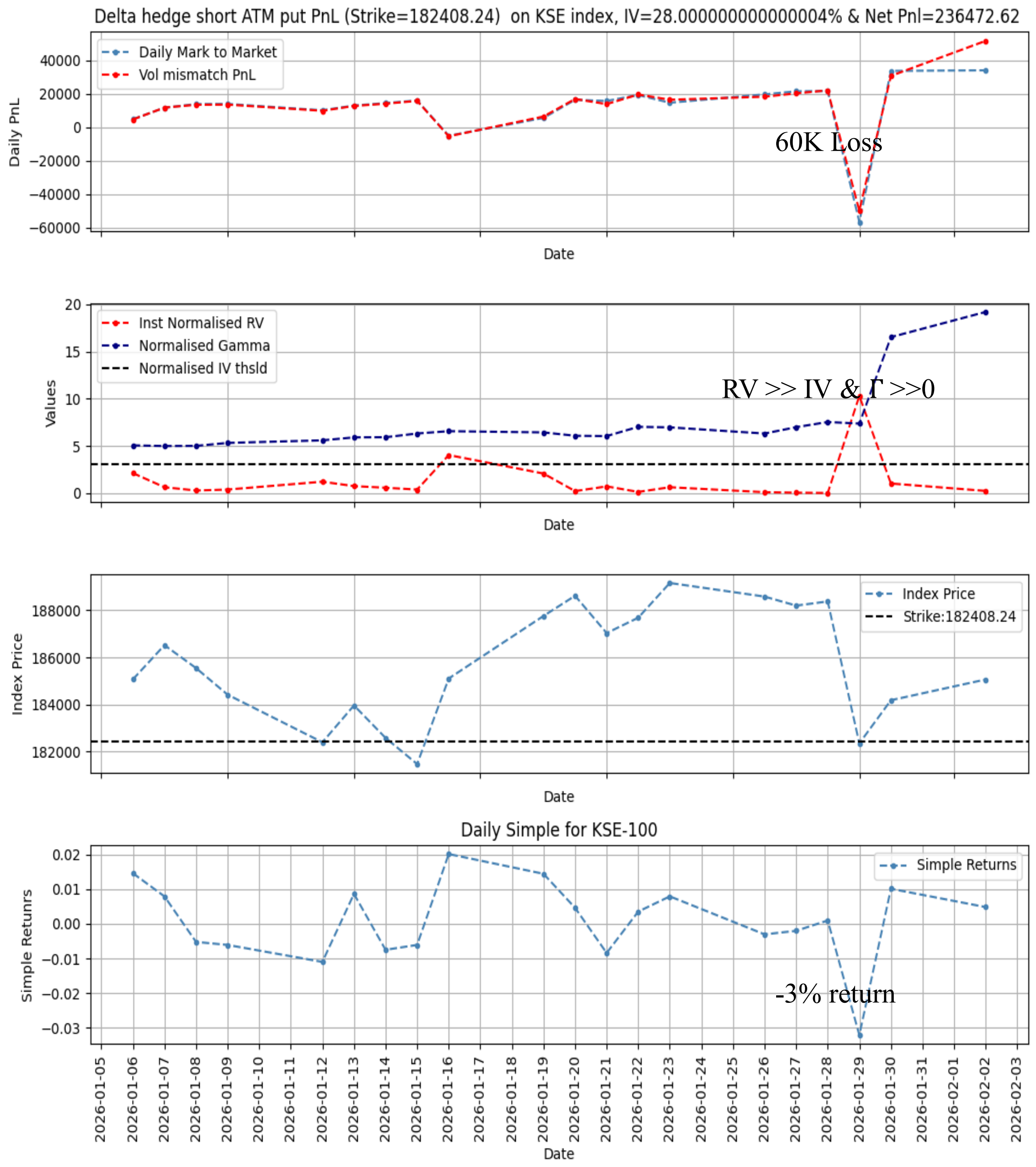
Choppy Markets & $\Gamma \gg 0$ Work Against Delta Hedged Option Seller

- Jumps in prices: time periods with higher instantaneous realized volatility result in loss for seller, provided **Gamma** takes a large value. More precisely short delta hedge position sees a *loss on t-th day's Mark to Market* if and only if:

$$\left(\frac{S_t - S_{t-1}}{S_{t-1}}\right)^2 > \sigma_{iv}^2\left(\frac{1}{252}\right) \quad \& \quad \Gamma \gg 0$$

- **Gamma is maximum around ATM,** this can be confirmed from the plot on right: on 29th of January 2026, index price is in the vicinity of option's strike K = 182408

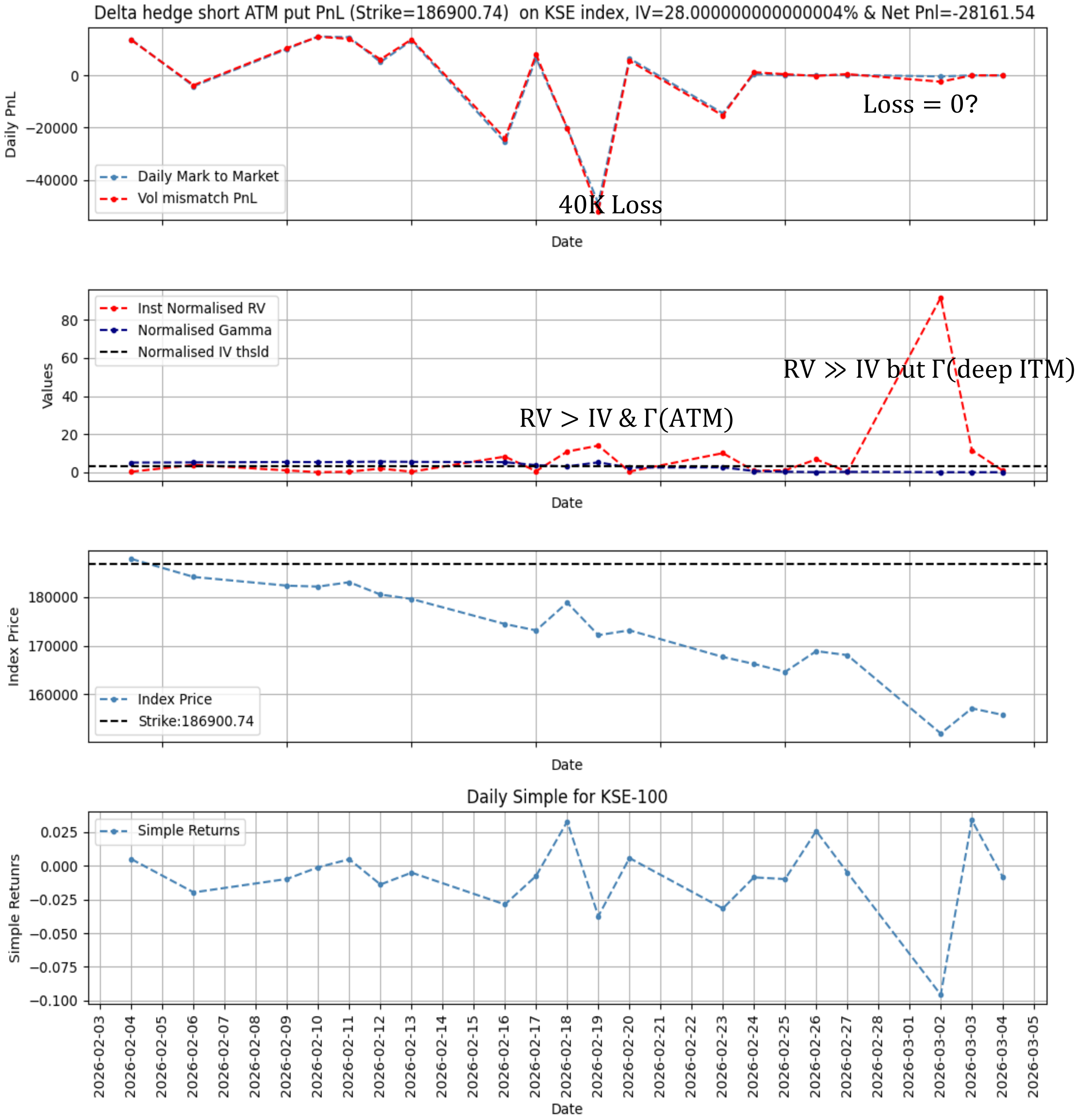
- Although short delta hedged put position had a profit of Rs. 236472 for Jan 2026 one-month put option priced with IV=28%, a 3% negative return was observed on 29th of January 2026 causing instantaneous realized volatility to cross implied volatility threshold. Moreover, stock was around Strike hence Gamma had a large value these factors add up to Rs. 50000 loss for short delta hedged put seller.



Choppy Markets & $\Gamma \gg 0$ Work Against Delta Hedged Option Seller

- Short delta hedged put position incurs a net loss in Feb-2026. Although instantaneous realized variance >> implied variance on 2nd March 2026, but index price is far away from the strike hence Gamma approximately zero and so is the PnL. Also for the scaling purposes Normalized Gamma = Index Price times Gamma.

- Hedging has the effect of reducing the variance of returns, but drastically increases transaction cost and introduces operational issues associated with position monitoring.



No intraday loss for IV>10%

- Overnight losses are main risk: which are upper bounded by: $50 \times \text{ClosePrice} \times \left(\sigma_{\text{IV}}^2 - \sigma_{\text{C-O}}^2\right)$
- Intraday losses are upper bounded by: $50 \times \text{Open Price} \times \left(\sigma_{\text{IV}}^2 - \sigma_{\text{O-C}}^2\right)$
- Total losses over a day are bounded by: $50 \times \sigma_{\text{IV}}^2 \times$ (Open Price $+$ Close Price) - 50$\times$ Open Price $\times \sigma_{\text{O-C}}^2$ $-50 \times$ Open Price $\times \sigma_{\text{O-C}}^2$

Break Even Implied Volatility & Fair Inception Prices

- **How to compute option prices on an underlying without options?** For instance: Compute 1 month 10% delta OTM Put implied volatility from index price history. A beautiful solution in the form of Break Even volatility was proposed by Bruno Dupire.

- Let current stock price be S_0 and K be the 10% delta put strike. Put seller collects the premium which is priced with some implied volatility σ_{IV}: $P_0 = \text{Premium} = \text{BSPutPrice}(S_0, K, T = \frac{20}{252}, r = 0, \sigma_{IV})$ and initially shorts $|\Delta_0^P| = 0.10$ shares of the underlying stock, so he is not exposed to any directional risk w.r.t underlying

- Next day when price of stock moves from $S_0 \rightarrow S_1$, put seller re-hedges to be delta neutral and incurs a *credit/debit* of $(|\Delta_1^P| - |\Delta_0^P|)S_1$, where recall that $\Delta_1^P = \text{BS\_Put\_Delta}(S_1, K, T = \frac{19}{252}, r = 0, \sigma_{IV})$. Along with initial credit of $P_{init} = P_0 + 0.10S_0$, delta hedging daily incurs a credit/debit roll = $(|\Delta_1^P| - |\Delta_0^P|)S_1 + (|\Delta_2^P| - |\Delta_1^P|)\, S_2 + \cdots . +(|\Delta_{19}^P| - |\Delta_{18}^P|)S_{19}$.

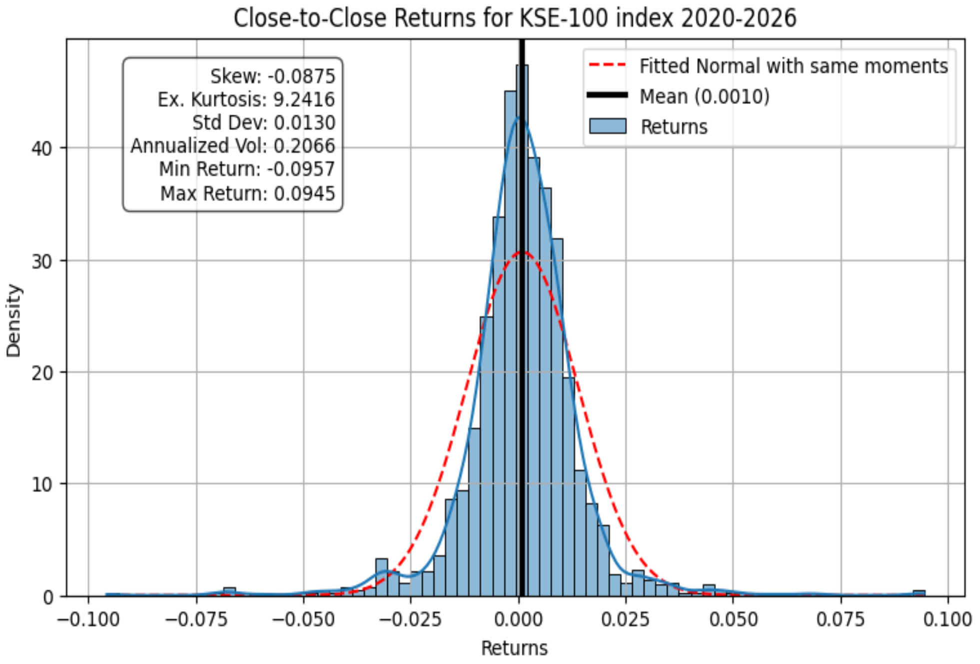
- On the day of maturity seller borrows back the shorted $|\Delta_{19}^P|$ shares at price S_{20} and covers the put claim – all liabilities covered. We can freely borrow/sell at 0 interest rate(justified r =0 for short terms); as we started from owning nothing and risking nothing, no arbitrage assertion imply that: $\text{BsPutPrice}(S_0, K, T = \frac{20}{252}, r = 0, \sigma_{IV}) + |\Delta_0^P|S_0 + (|\Delta_1^P| - |\Delta_0^P|)S_1 + (|\Delta_2^P| - |\Delta_1^P|)\, S_2 + \cdots . +(|\Delta_{19}^P| - |\Delta_{18}^P|)S_{19} - |\Delta_{19}^P|S_{20} - \max(K - S_{20}, 0) = 0$

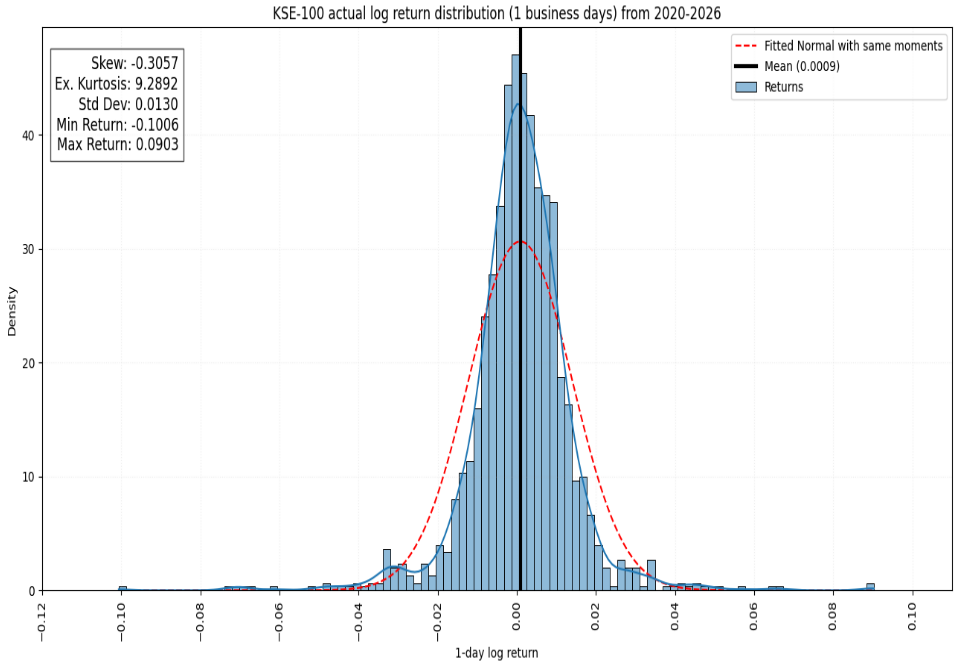
- Equivalently said: given initial spot, length of vanilla option contract and delta corresponding to strike K, no arbitrage restriction asserts that at least in expectation $\sigma_{IV} = \sigma_{IV}(\Delta, T)$ satisfy:

$$\sum_{t=1}^{T} \frac{\Gamma_{t-1} S_{t-1}^2}{2} \left(\sigma_{iv}^2 (1/252) - \left(\frac{S_t - S_{t-1}}{S_{t-1}} \right)^2 \right) = 0$$

Market Crash of 1987 and Introduction of Index Volatility Skew

- Before 1987, all the options on indexes regardless of maturity date and strike were priced with same volatility parameter(as PSX has proposed) i.e., strike/time to maturity vs implied volatility graph was flat.
- The volatility skew anomaly was discovered after the stock market crash of October 1987 and left many practitioners and academics wondering why this shape of the implied volatility over strike prices existed. Ever since the October 1987 market crash the volatility skew in indices, that still haunts the minds of traders, has served as a reminding grin of the historical devastating effect of naively using historical volatility while pricing options
- We can see from the return's histogram that indexes go through larger downside moves compared to upside moves(negative skew) and this increases the demand of deep OTM puts by diversified long equity portfolios.
- **Crash-O-Phobia:** An index is basked of stocks and correlation between those stocks increases in crash ; magnifying the effective volatility, exactly when deep OTM puts payoff.
- Although individual stocks have a binary event risk(earnings, biotech trials, take-overs) large positive returns do happen but for that to show up as large return in index, all those positive events for stocks making up index must happen at the same time(likelihood almost 0)





Implied Volatility Skew Premium In Index Options

- For each given maturity and initial delta, we iteratively solve for the Break-Even implied volatility satisfying:

$$\sum_{t=1}^{T} \frac{\Gamma_{t-1} S_{t-1}^2}{2} \left(\sigma_{iv}^2 (1/252) - \left(\frac{S_t - S_{t-1}}{S_{t-1}} \right)^2 \right) = 0$$

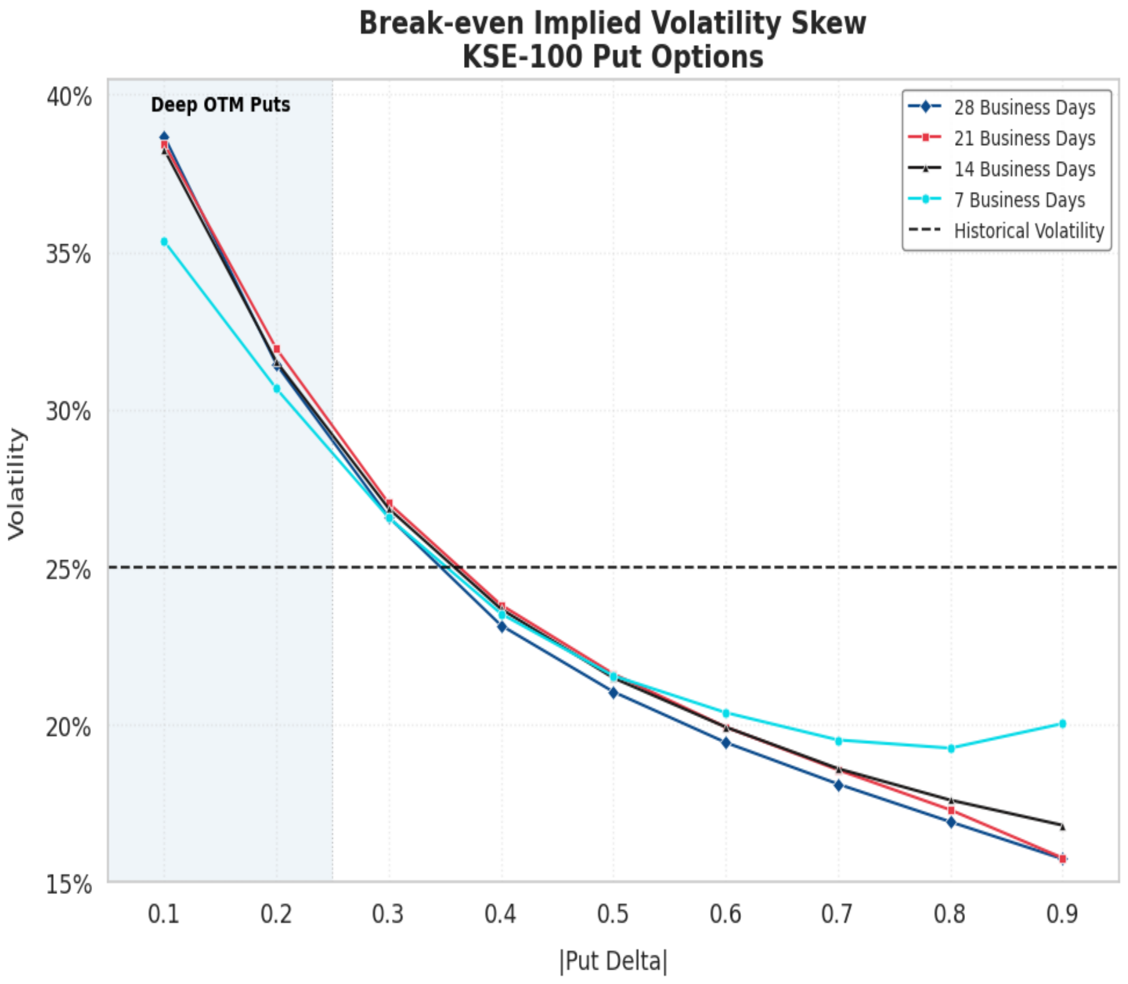
on historical KSE-100 index data to get a fair Skew.

- **IV Skew Premium** reflects how much more expensive downside protection (OTM puts) is relative to upside exposure (OTM calls).

- Equity markets consistently price crash risk into puts, creating a *negative skew* (higher IV for lower strikes) in index options. A positive skew premium indicates market fear of downside

$$\text{Skew Premium} = \text{IV}_{|\text{Put Delta}|=0.25} - \text{IV}_{\text{Call}=0.25}$$

- Since Call Delta – Put Delta =1, we can see extract volatility skew from the Break-even implied vol skew on KSE-100 Options



Tenor(Business Days)	Volatility Skew Premium
28	11.5
21	11
14	10.8
7	9

Iterative Method To Find Break-Even Implied Volatility(Sticky Delta)

- **Sticky delta rule:** popular heuristic that attempts to describe the movement of the implied volatility smile as a function of underlying's moves; it says that smile moves with the underlying so as to keep
- Fix $\Delta := \Delta_C^{BS} := \frac{\partial C}{\partial S} = N(d_1)$ from $[0.1, 0.2, \ldots, 0.8, 0.9]$; Δ being a proxy for probability that option expires in the money. Under sticky delta model implied volatility is a function of Δ.
- S_{i-1} be the initial spot price corresponding to i^{th} moving average window. With initial guess on implied volatility over 21 business day $\widehat{\sigma_0}$ (take 21 day realized vol) and solve for strike $\widehat{K_{i-1}}$ that satisfies

$$\Delta = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{\ln\left(\frac{S_{i-1}}{\widehat{K_{i-1}}}\right) + \left(\frac{\sigma_0{}^2}{2}\right)\frac{21}{252}}{\sigma_0\sqrt{\frac{21}{252}}}} e^{-\frac{z^2}{2}} dz$$

- Short a 21 day call option with premium satisfying BSM call price at strike $\widehat{K_{i-1}}$, initial spot S_{i-1} and implied vol$=\widehat{\sigma_0}$, also hedge by going long Δ shares of the underlying. Every day, delta hedge and consequently realize total P&L of $\sum_{t=i-1}^{t=i+19} \frac{\Gamma_{t-1} S_{t-1}^2}{2}\left(\widehat{\sigma_0}{}^2 (1/252) - \left(\frac{S_t - S_{t-1}}{S_{t-1}}\right)^2\right)$.
- $$\frac{1}{481}\sum_{i=1}^{481}\sum_{t=i-1}^{i+19} \frac{\Gamma_{t-1} S_{t-1}^2}{2}\left(\sigma_{iv}^2 (1/252) - \left(\frac{S_t - S_{t-1}}{S_{t-1}}\right)^2\right) = 0$$

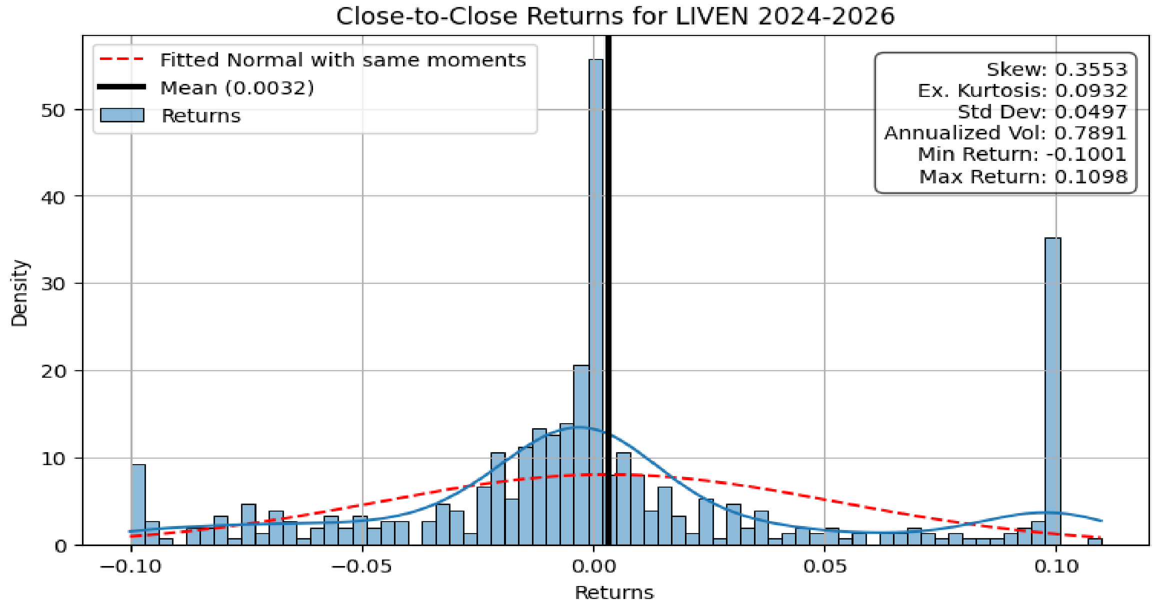
$$\boxed{S_0, S_1, S_2, S_3 \ldots, S_{20}}, S_{21}, S_{22}, \ldots\ldots, S_{500}$$

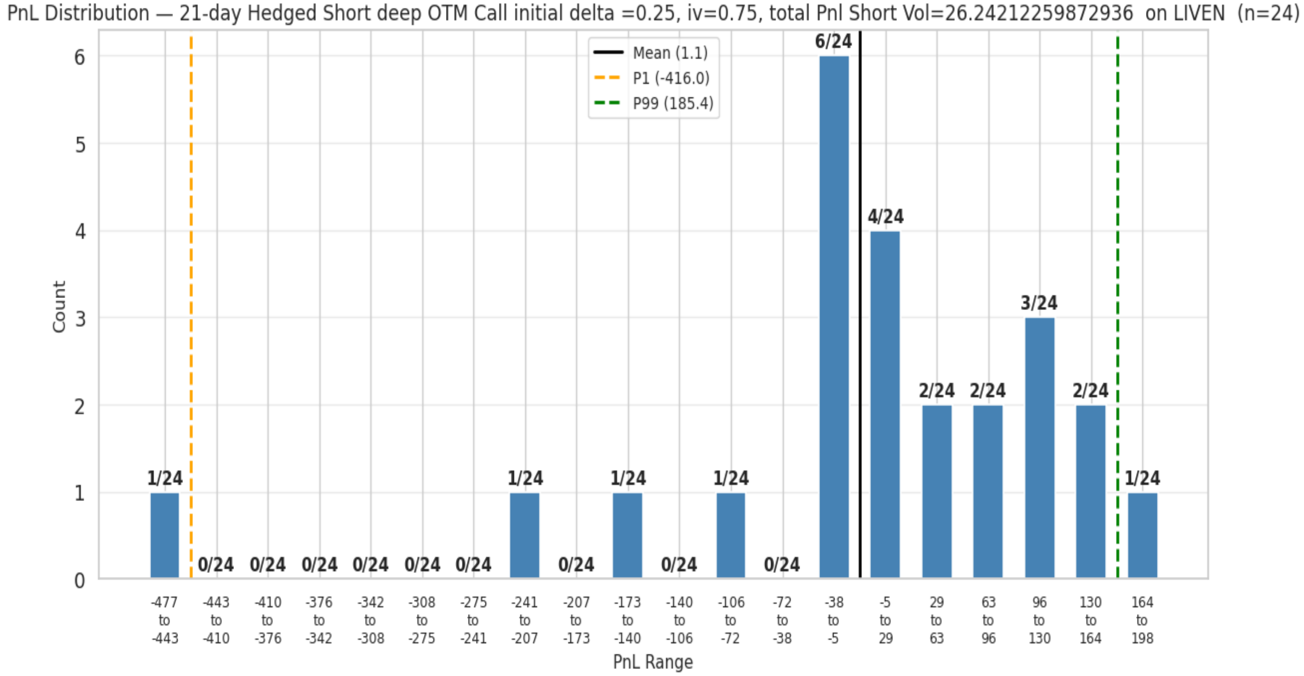
$$S_0, \boxed{S_1, S_2, S_3 \ldots, S_{20}, S_{21}}, S_{22}, \ldots\ldots, S_{500}$$

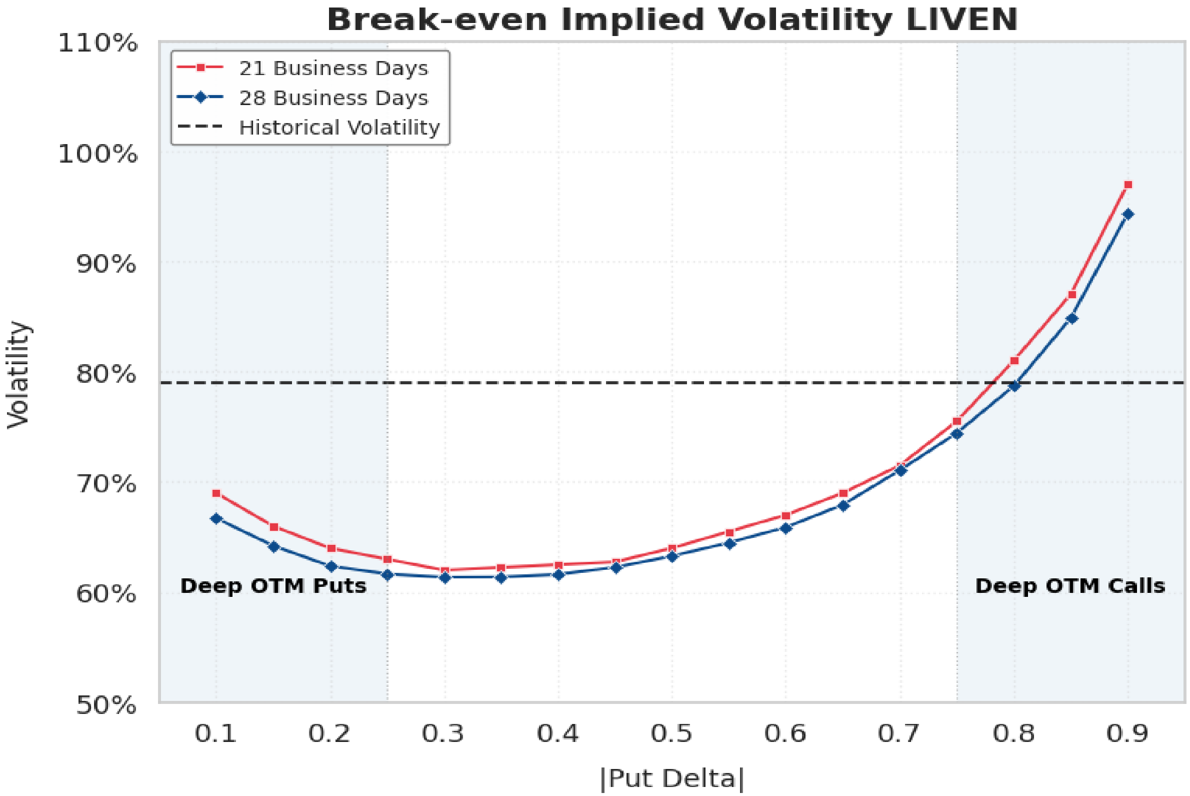
$$S_0, S_1, \boxed{S_2, S_3 \ldots, S_{20}, S_{21}, S_{22}}, \ldots\ldots, S_{500}$$

Single Stock Options: Volatility Smile

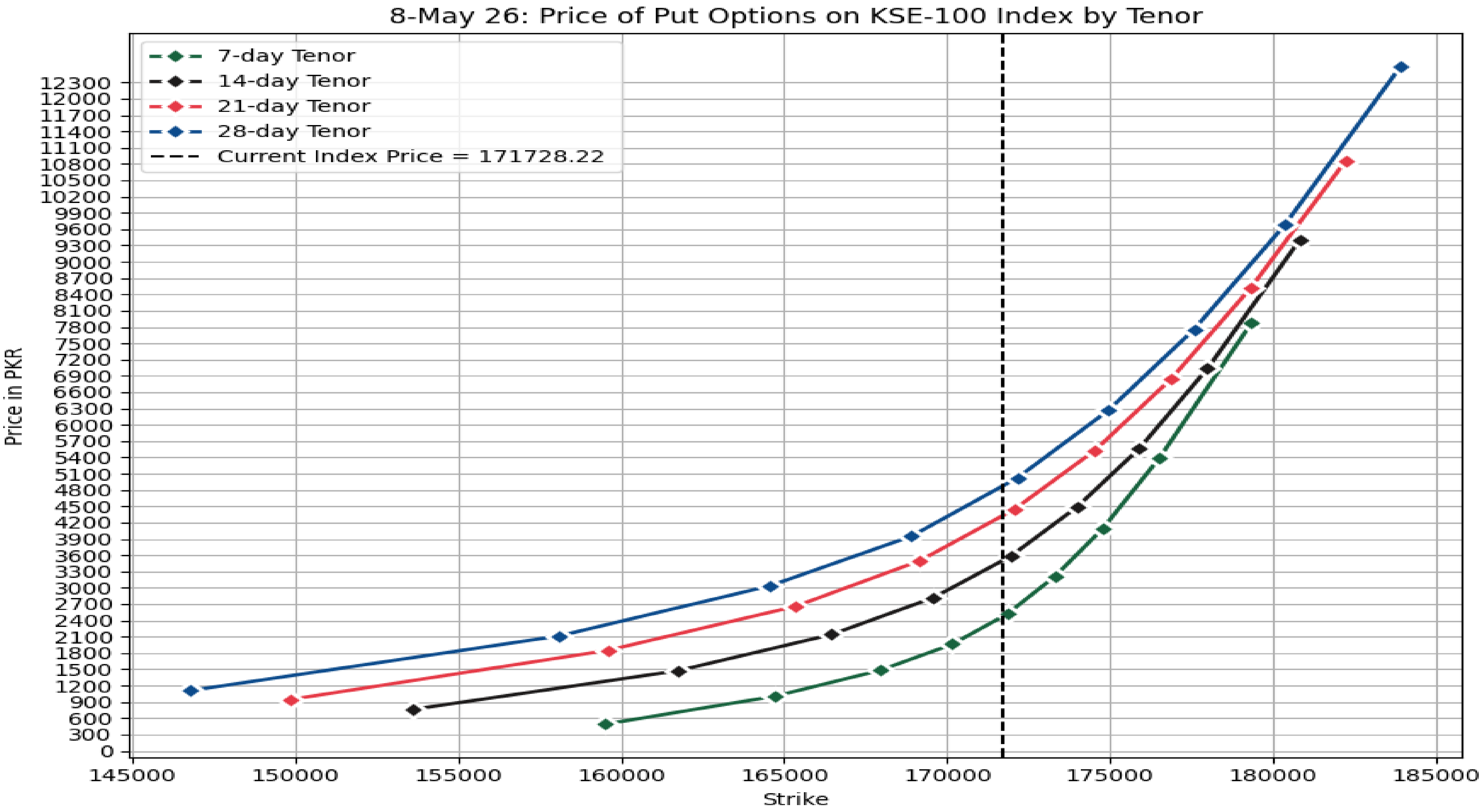
- For stocks large moves happen in **both directions** and returns are **fat tails** and : which means that far out-of-the-money options are a lot more likely to pay-off than would be the case if the distribution of log returns was normal(as assumed by BSM). So, they are more valuable. The only way BSM model can account for this is to put in a higher implied volatility giving us a smile in single stock options.
- LIVEN has a positive skew of 0.35 which means that larger upside moves compared to downside moves, so deep OTM calls must be more expensive than deep OTM puts.



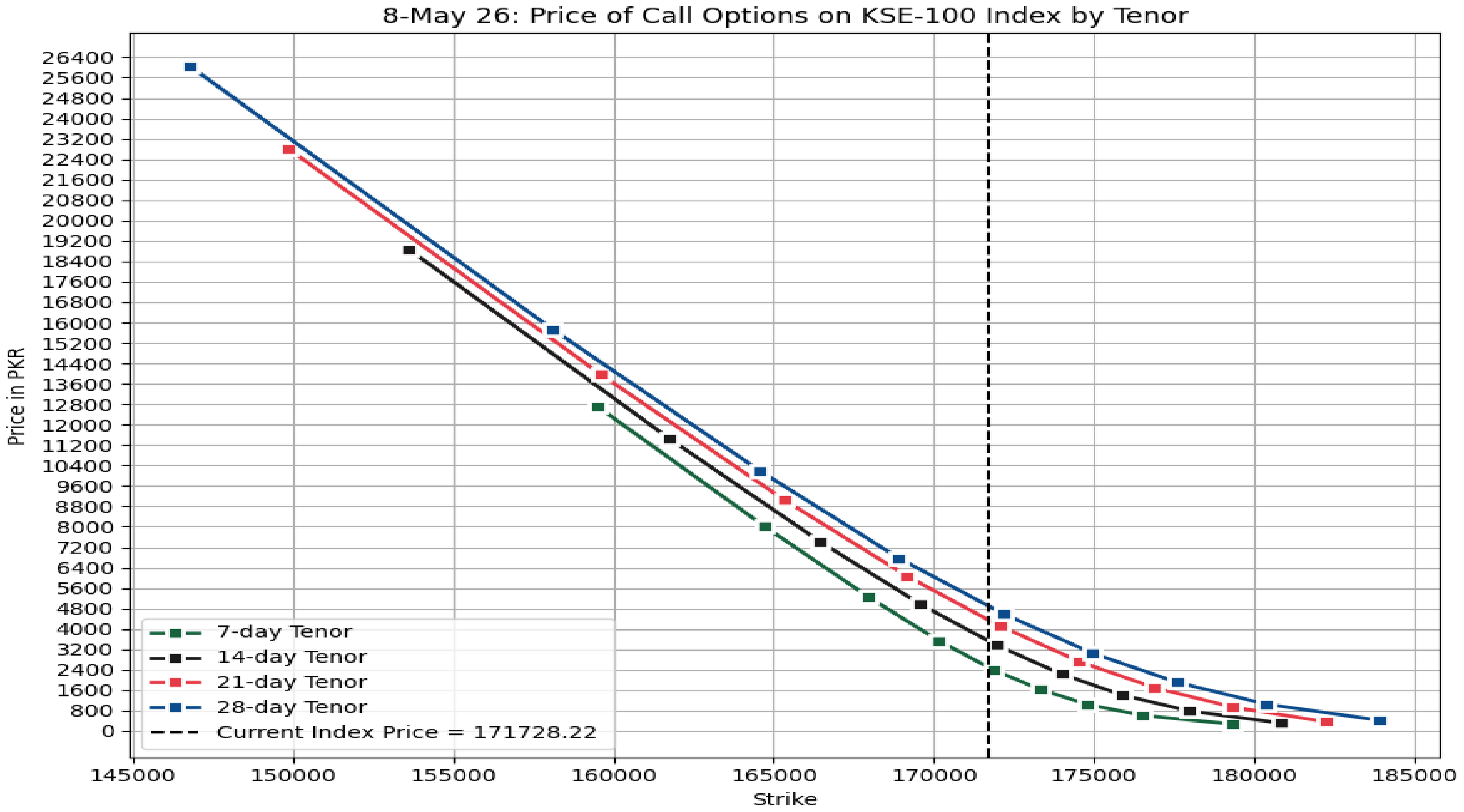




Arbitrage Free Price For KSE-100 Index Put Options



Price For KSE-100 Index Call Options(Put-Call Parity Holds)

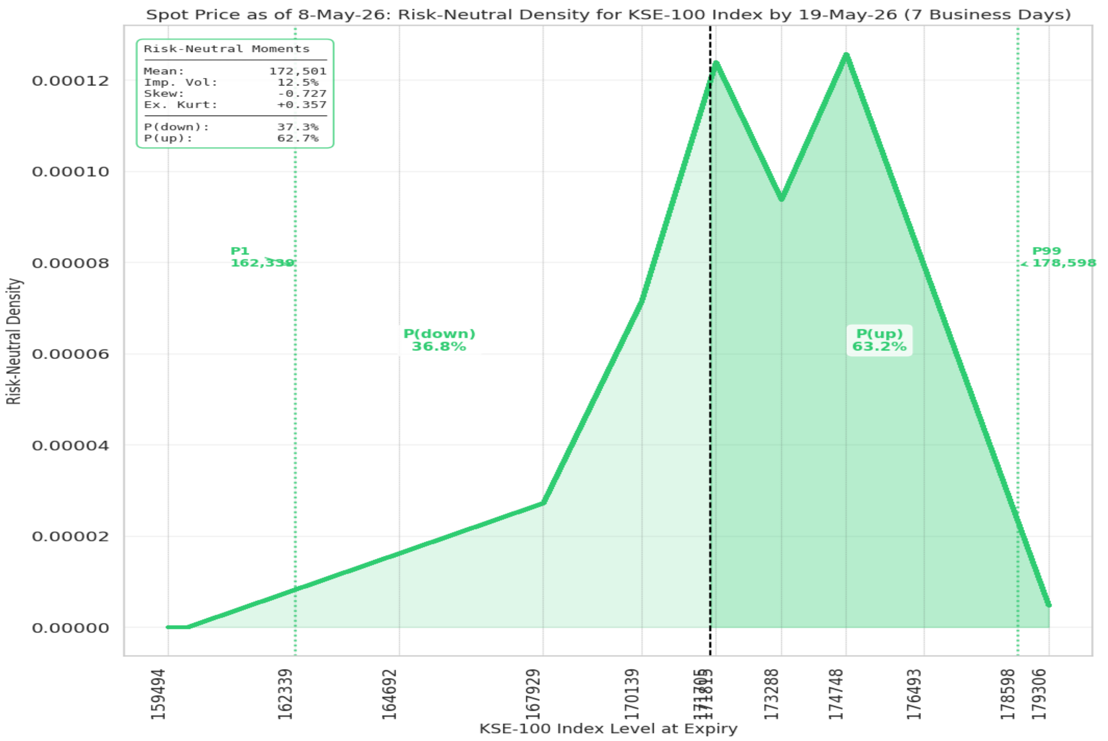


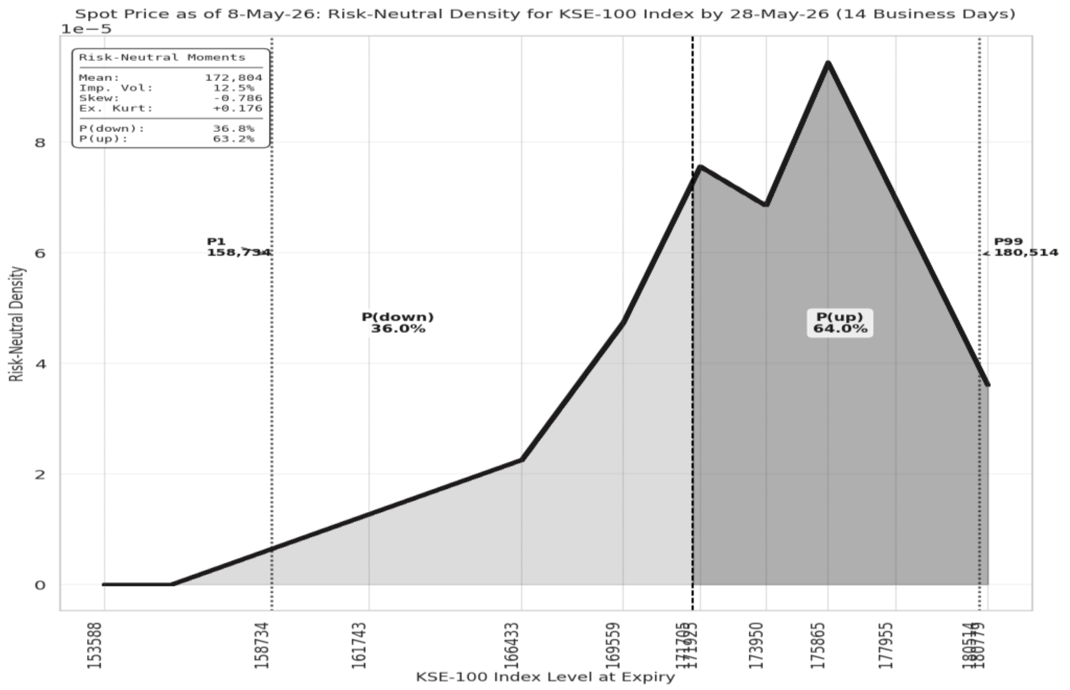
Predicting Future Index Level Via Option Prices

- **Breeden-Litzenberger Formula** Given call prices C(K) as a function of strike K, the risk-neutral density is simply the second derivative of the call price with respect to strike

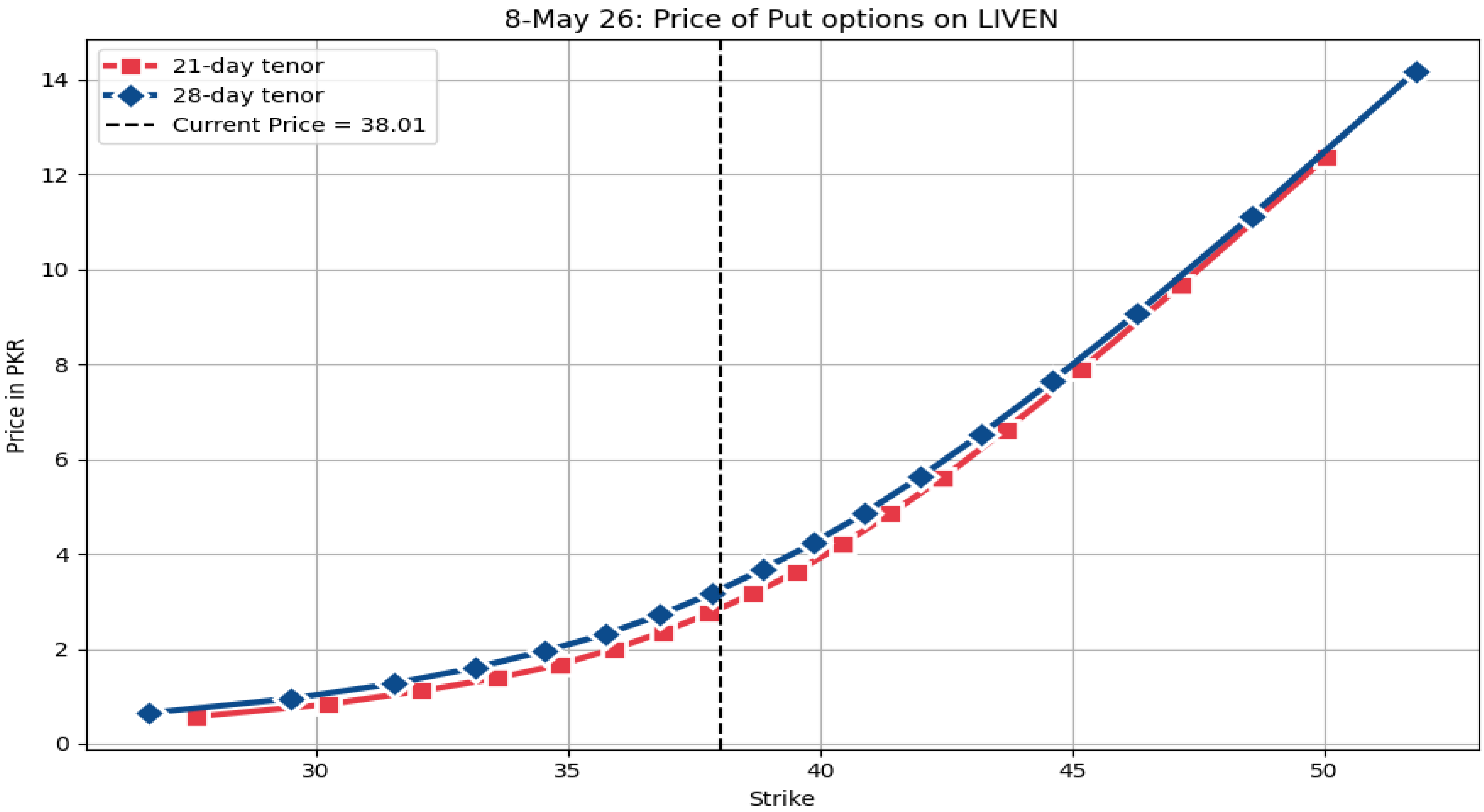
$$\mathbb{P}_{\mathbb{Q}}(S_T = K|S_0 = 171728) = e^{rT}\frac{\partial^2 C}{\partial K^2}(T, K) \approx \frac{\partial^2 C}{\partial K^2}(T, K)$$

- With a probability of 0.632 under RN measure, returns over next 7 business days will be positive but 99th percentile of the returns is $R_{99}^{IMP}(T = 7, S_0) \times 100 = \frac{178598-171728}{171728}$=4%, but negative return corresponding to 1st percentile is $R_1^{IMP}(T = 7, S_0) \times 100 = \frac{171728-162330}{171728}$=5.5%
- With a probability of 0.635 under RN measure, returns over next 21 business days will be positive but 99th percentile of the returns is $R_{99}^{IMP}(T = 21, S_0) \times 100 = \frac{181983-171728}{171728} \approx 6\%$, but negative return corresponding to 1st percentile is $R_1^{IMP}(T = 21, S_0) \times 100 = \frac{171728-157028}{171728} \approx 8.6\%$. $R_{99}^{IMP}(T = 28, S_0) \times 100 = \frac{183654-171728}{171728} \approx 7\%$, whereas $R_1^{IMP}(T = 28, S_0) \times 100 = \frac{171728-149850}{171728} \approx 12.7\%$

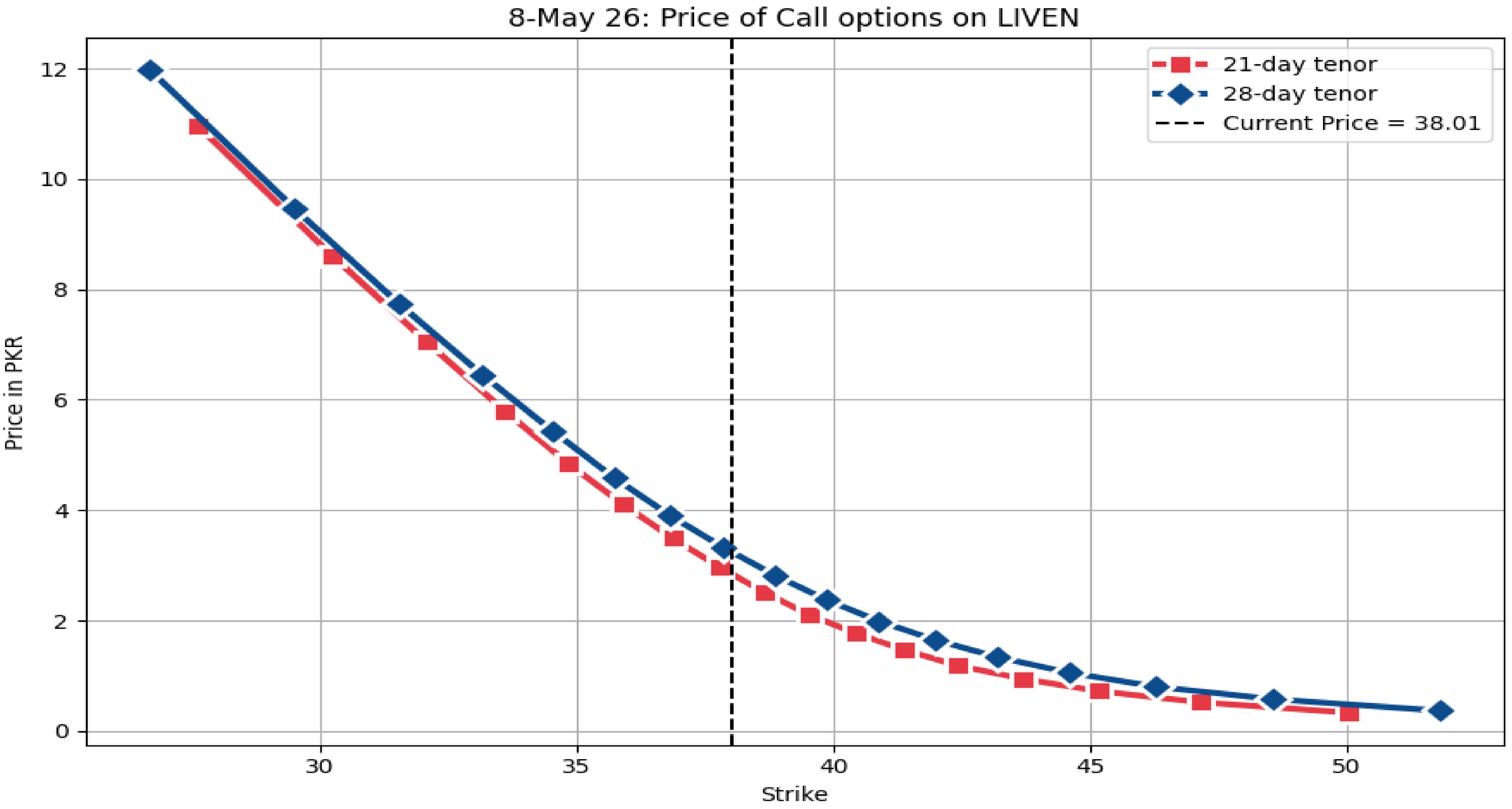




Arbitrage Free Price For Single Stock Put Options: LIVEN

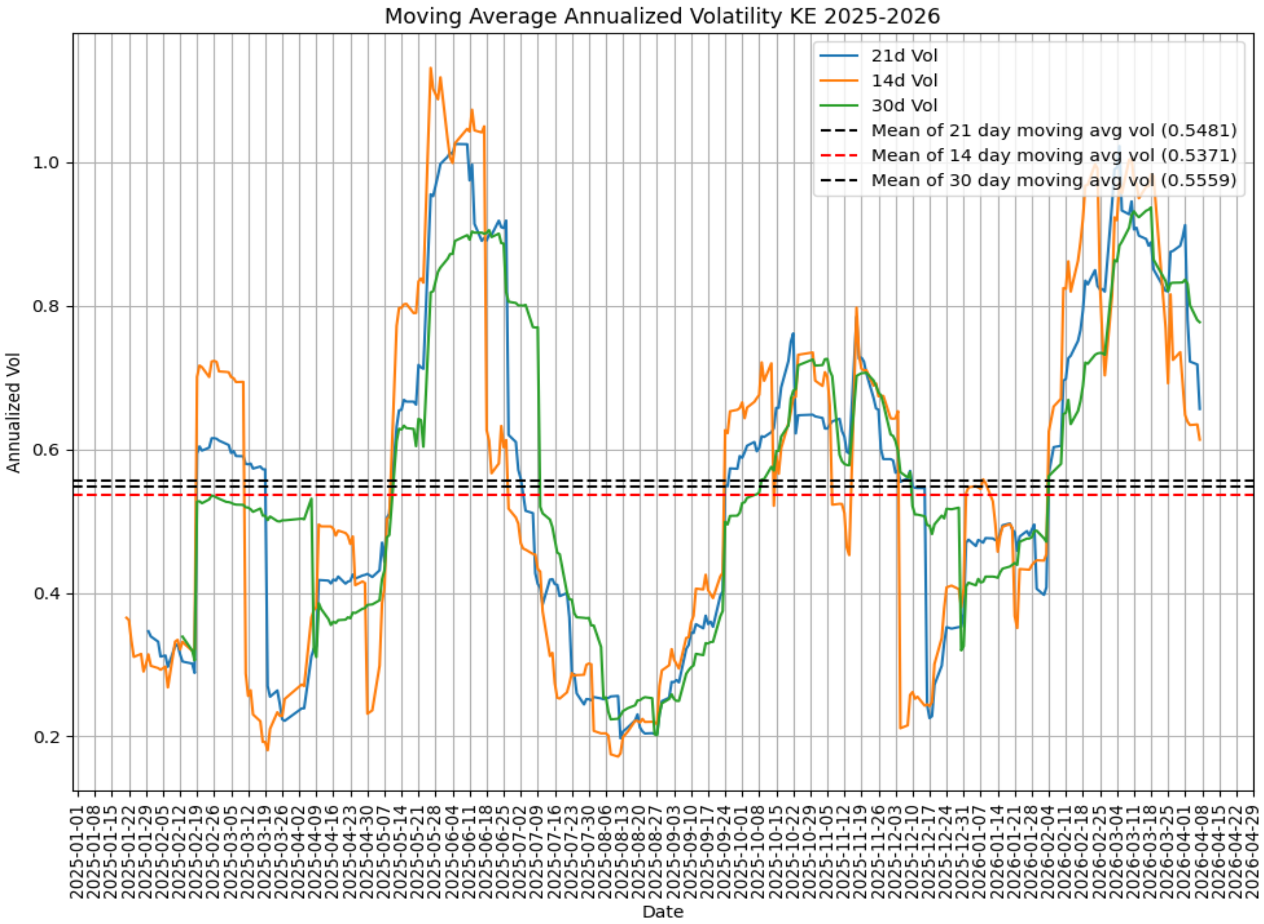


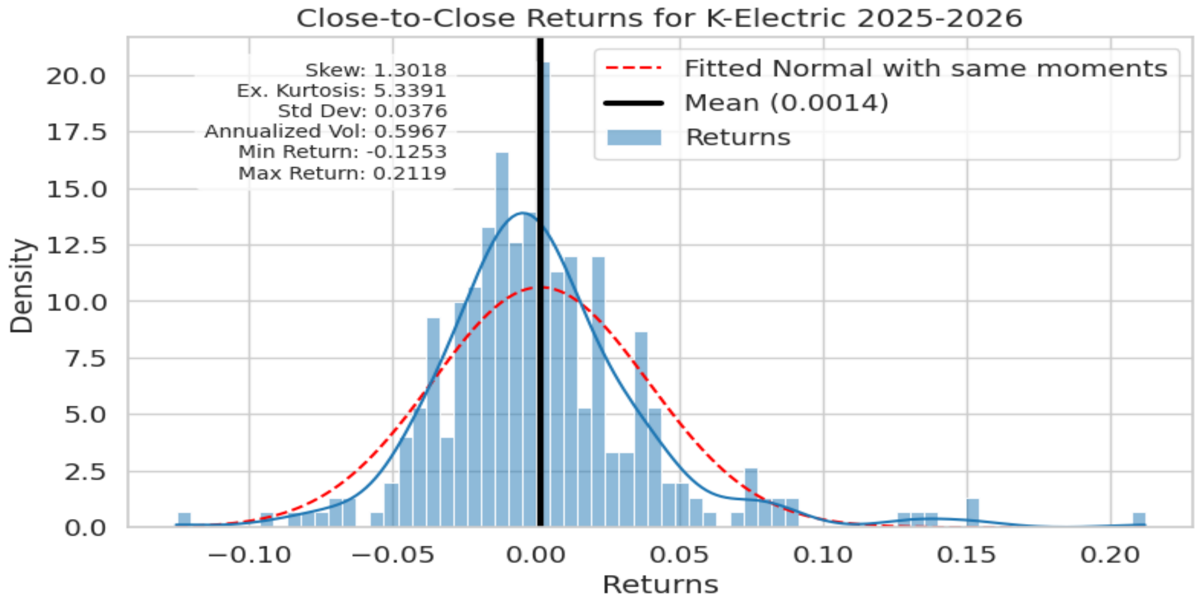
Call Option Price On LIVEN(Put-Call Parity Holds)

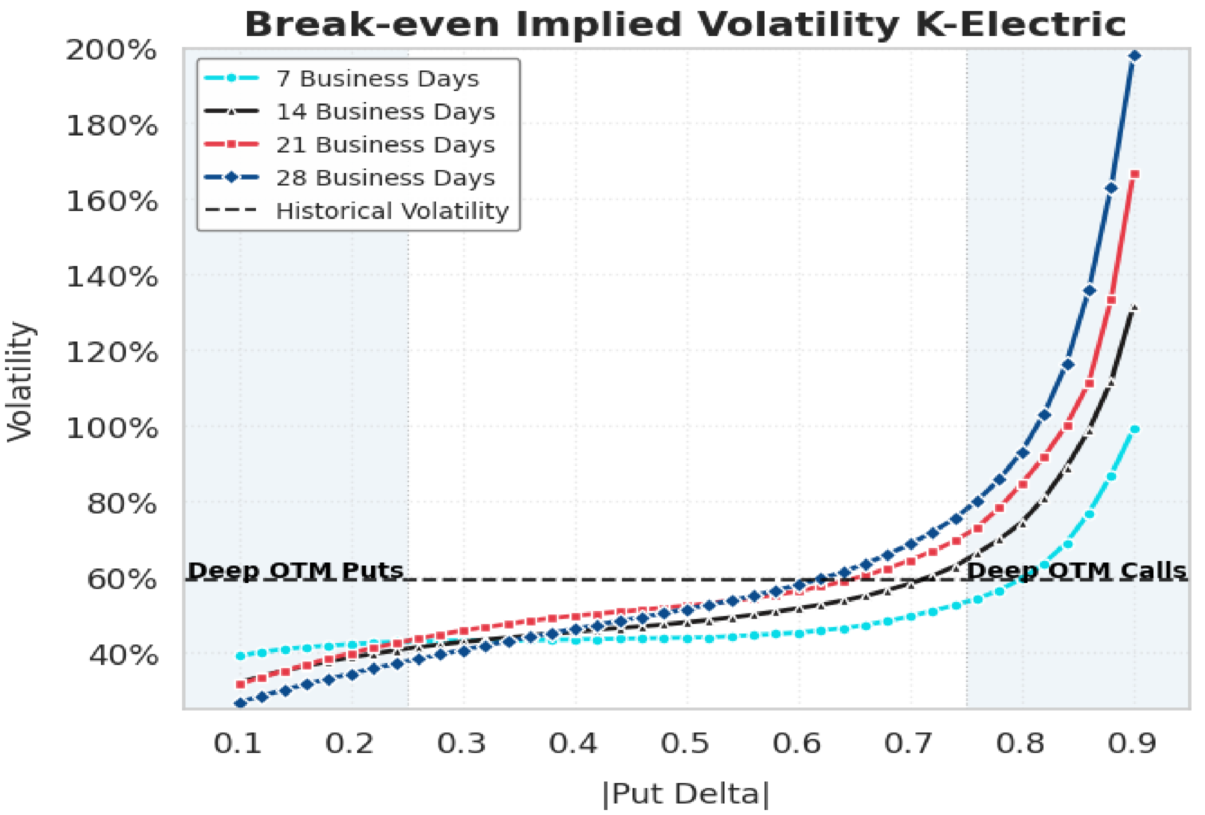


Single Stock Options: IV vs RV

- K-Electric has a more pronounced positive skew of 1.3018 compared to LIVEN; deep OTM calls must be priced much higher than deep OTM puts.
- Max of 14 day realized volatility is 115% and deep OTM calls with 14 day tenors are priced at 130% implied volatility.





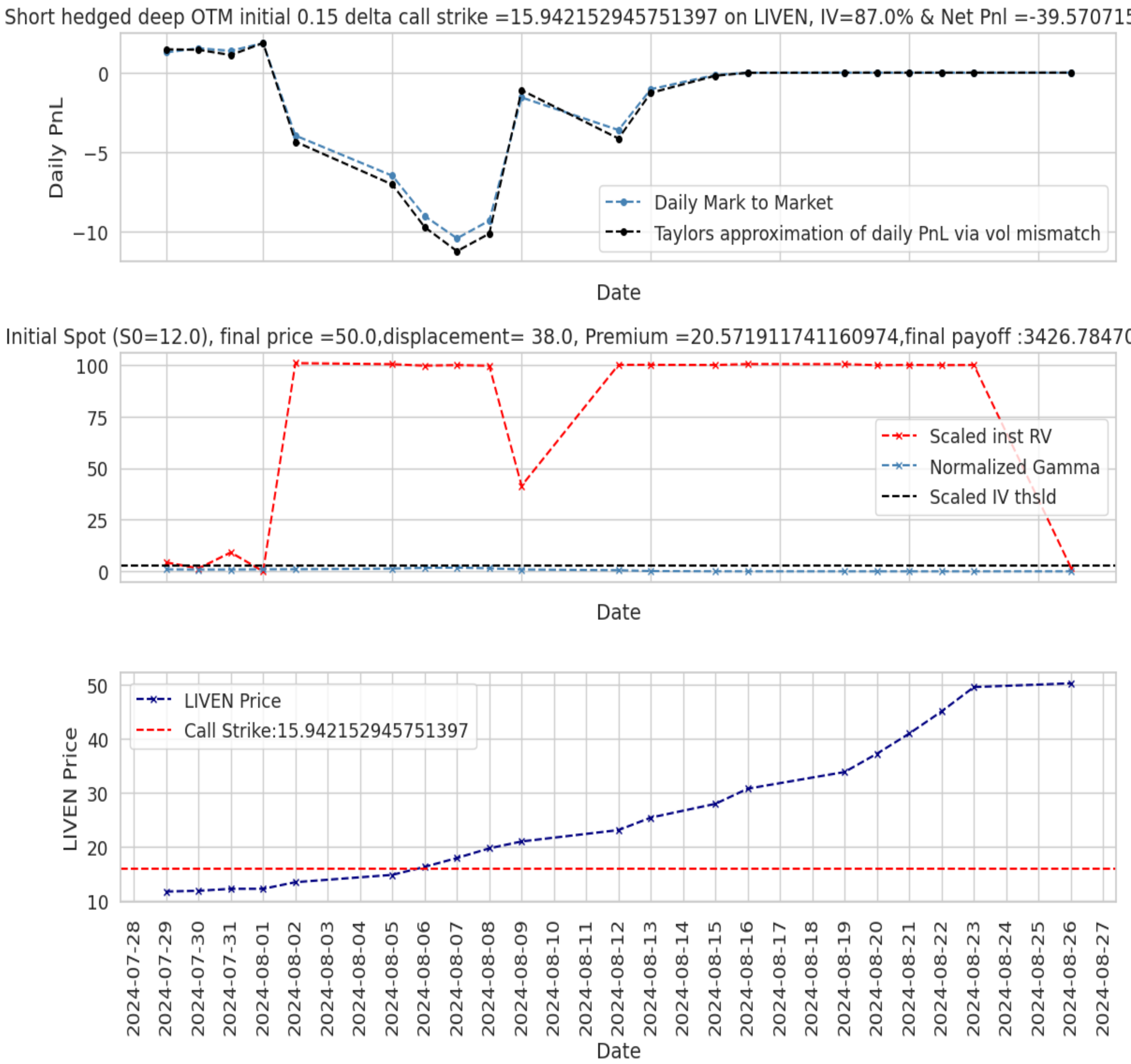


Single Stock Deep OTM Call Options: Lottery Ticket

- Deep OTM calls are like lottery tickets for retail investors; 0.15 delta call(28th July-2024 to 27th Aug-2024) on LIVEN costed 0.257 and returned 34.3 which is 13000%. Delta hedging saved the seller.

PnL Statistics — 21-day Hedged Short initial 0.15 delta Call

Statistic	Value
Samples	24.00
Mean	1.95
Std Dev	124.78
Skewness	-2.61
Kurtosis (exc)	7.56
P1	-424.25
P25	-3.60
P75	77.32
P99	125.28
Min	-490.79
Max	129.76
Total PnL	46.83



Black Scholes Pricing & Greeks

- K be the strike, S being the current spot price, T be the time remaining to maturity in years, r being the annualized risk-free rate and σ_{iv} be the annualized implied volatility. Furthermore, if we let $d_1 := \frac{\ln\left(\frac{S}{K}\right)+\left(r+\frac{\sigma_{iv}^2}{2}\right)T}{\sigma_{iv}\sqrt{T}}$, $d_2 = d_1 - \sigma_{iv}\sqrt{T}$, $N(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{z^2}{2}}dz$ and $n(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$, and assume no-dividends we have the following **Black Scholes Call Price & Put Price** :

$$C^{BS}(K,S,T,\sigma_{iv}) = SN(d_1) - Ke^{-rT}N(d_2), \quad P^{BS}(K,S,T,\sigma_{iv}) = C^{BS}(K,S,T,\sigma_{iv}) - S + Ke^{-rT}$$

i.e., *long call is essentially long $N(d_1)$ shares of underlying and borrowing cash amount of $Ke^{-rT}N(d_2)$*

- **Delta:** Sensitivity of call price to changes in the spot price $\Delta_C^{BS} := \frac{\partial C}{\partial S} = N(d_1)$ *and often referred to as a proxy to probability of option expiring in the money, delta has a smaller value if $\ln\left(\frac{S}{K}\right) < 0$ i.e., at the moment call option is OTM.. Put delta satisfies* $\Delta_P^{BS} = \Delta_C^{BS} - 1$.

- **Gamma:** Sensitivity of delta to changes in the spot price , $\Gamma := \frac{\partial \Delta^{BS}}{\partial S} = \frac{n(d_1)}{S\sigma_{iv}\sqrt{T}} = \frac{e^{-\frac{d_1^2}{2}}}{S\sigma_{iv}\sqrt{2\pi T}}$. For ATM options, $\Gamma^{-1} = S\sigma_{iv}\sqrt{2\pi T}e^{\frac{\sigma_{iv}^{-2}\left(r+\frac{\sigma_{iv}^2}{2}\right)^2 T}{2}}$ i.e., Gamma explodes to infinity for ATM options, and for ITM and OTM options it converges to 0, as they approach to maturity.

Remark: Gamma and Vega are same for call and put

- **Vega:** Partial derivative of the option value with respect to implied volatility. If the options are priced using BSM model, the call Vega is given by the expression Vega $:= \frac{\partial C}{\partial \sigma_{iv}} = \Gamma\, S^2\sigma_{iv}\, T = \frac{n(d_1)S^2\sigma_{iv}\, T}{S\sigma_{iv}\sqrt{T}} = Sn(d_1)\sqrt{T}$ i.e., sensitivity to volatility is proportional to time to maturity. This is slightly different in character from the Greeks discussed so far; as they were derivatives with respect to variables, whereas Vega involves taking derivative with respect to a parameter. If the BSM formalism was correct, the implied volatility would be a constant and this Greek could not even be defined. However, among traders it's an extremely important risk number.

Risk Comparison: Strangle vs Straddle

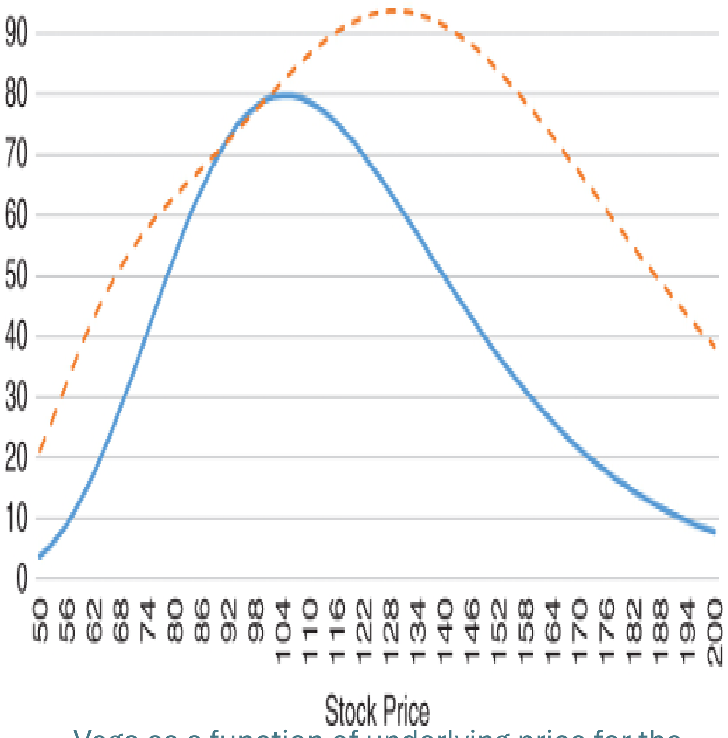
- Current Spot is \$100. Contract A: short an (ATM straddle)$\times$ 100 expiring in one year from now and Contract B: shorts (70/130 strangle)$\times$ 170 which position is risker and demands higher margins? Both contracts have same initial exposure to changes in implied volatility i.e., same initial Vega Risk. Short positions are prone to loss as stock moves far away from \$100. Consequently, Vega Risk of both contract changes:
- Straddle and Strangles without hedging expected: $\text{PnL} = \text{Vega}(\sigma_{IV} - \sigma_{RV})$, if broker sales Contract A and Contract B at an implied volatility of 30% but market realizes 70%:

Average	−\$3,230
Standard deviation	\$9,190
Skewness	−4.2
Excess kurtosis	21.7
Median	−\$1,650
10th percentile	−\$10,085
Minimum	−\$197,526
Percent profitable	36%

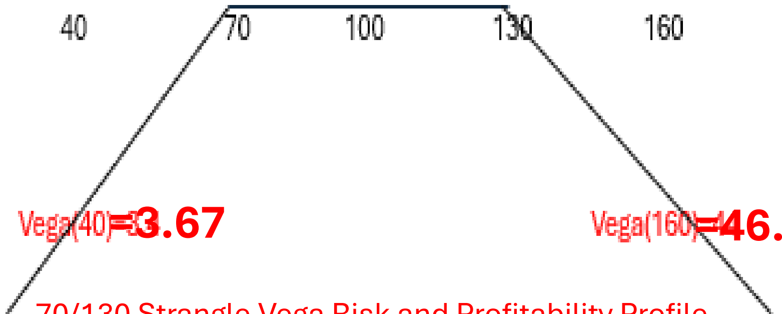
Summary and Stats for mispriced short strangle

Average	−\$3,071
Standard deviation	\$5,425
Skewness	−4.31
Excess kurtosis	29.01
Median	−\$2,143
10th percentile	−\$7,069
Minimum	−\$94,084
Percent profitable	25%

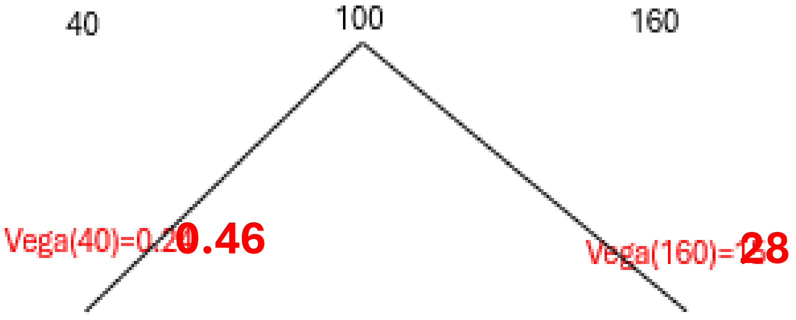
Summary and Stats for mispriced short straddle



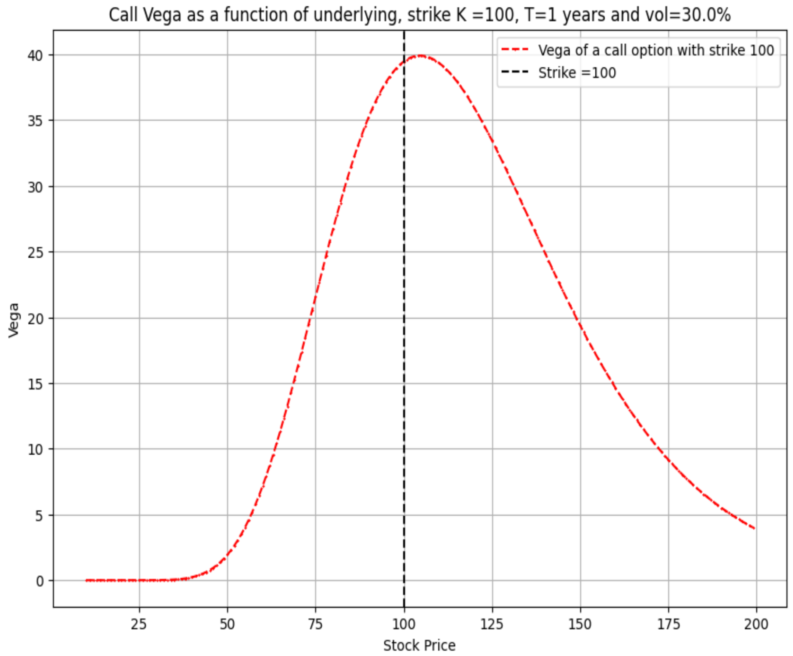
Vega as a function of underlying price for the straddle (sold line) and 70/130 strangle (dashed line)



70/130 Strangle Vega Risk and Profitability Profile



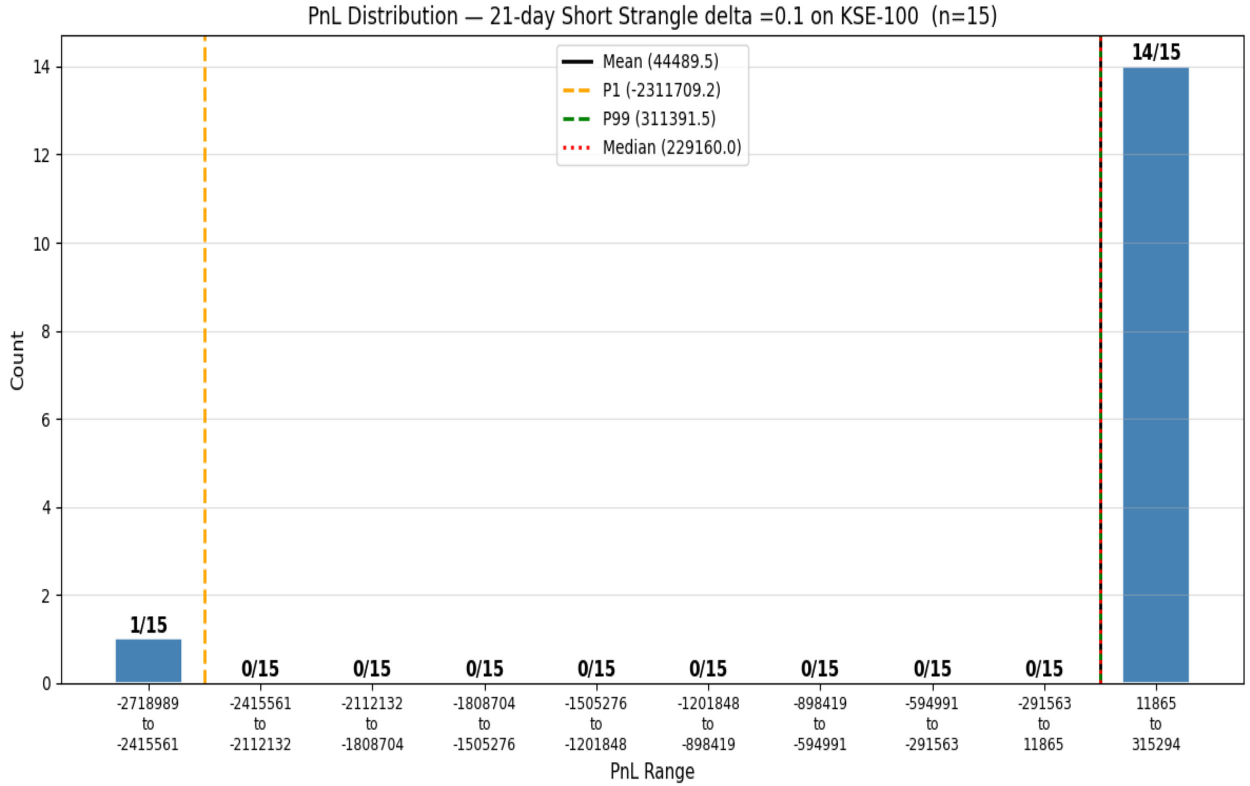
ATM Straddle Risk and Profitability Profile

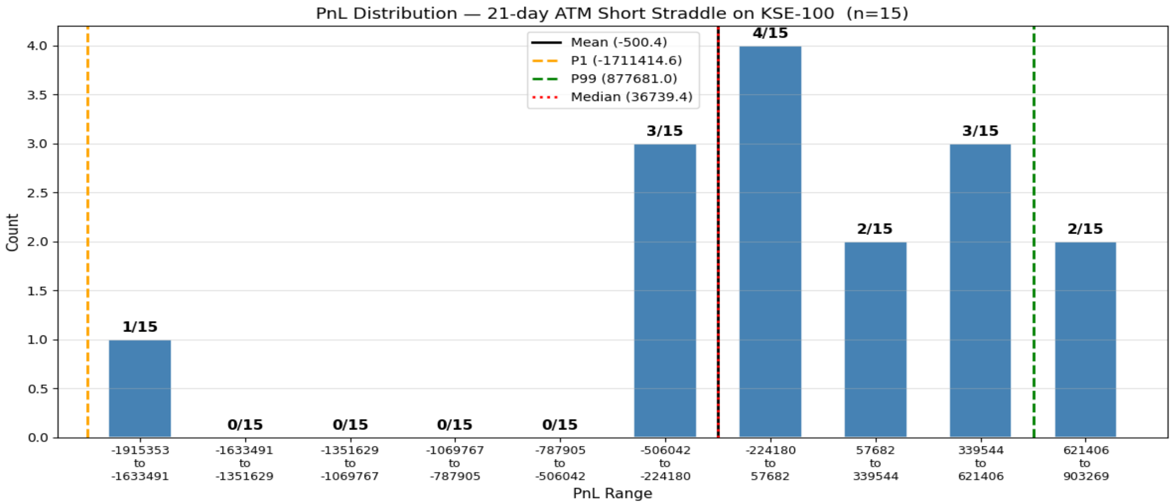


Strangle vs Straddle on KSE-100

- Deep OTM short strangle: 0.10 delta call and -0.10 delta put makes money 14/15 times and loses all the fortune when index plummeted in Feb almost Rs. 3 million loss. Almost ATM short strangle(0.45 delta) behaves like ATM short straddle.

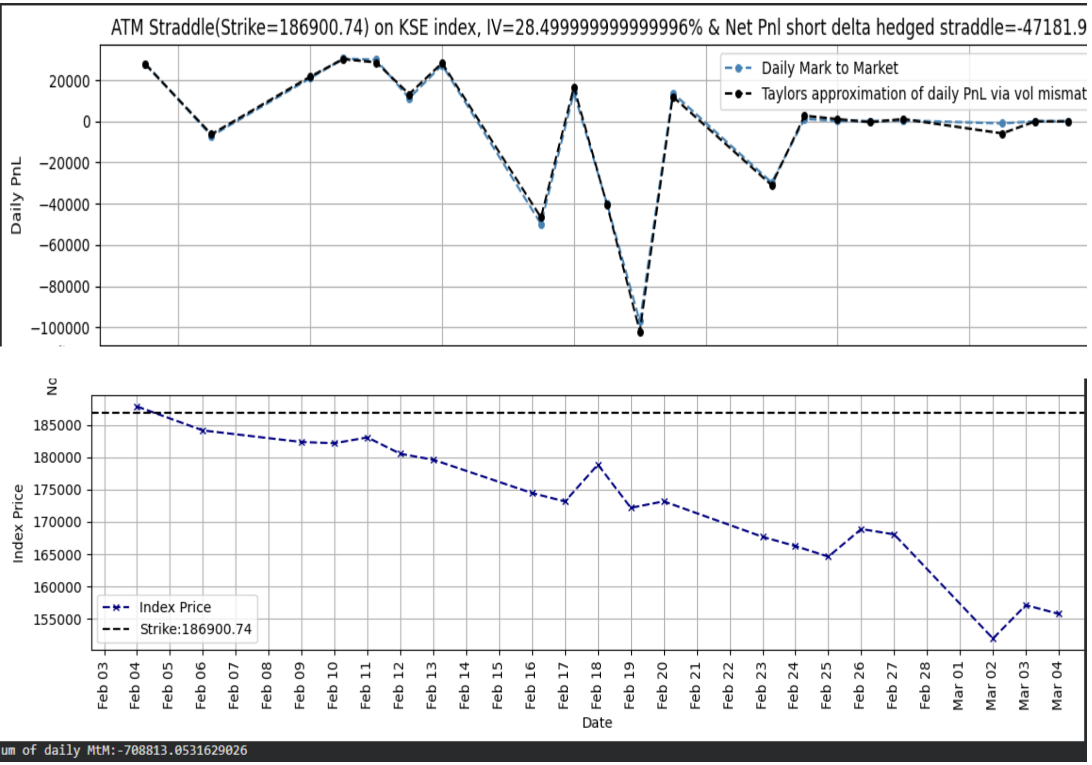
PnL Statistics — 21-day ATM Short Straddle		PnL Statistics — 21-day 0.1 delta Short Strangle	
Statistic	Value	Statistic	Value
Samples	15.00	Samples	15.00
Mean	-500.36	Mean	44489.54
Std Dev	655598.66	Std Dev	739680.48
Median	36739.37	Median	229160.00
Skewness	-1.34	Skewness	-3.46
Kurtosis (exc)	2.32	Kurtosis (exc)	9.99
P1	-1711414.55	P1	-2311709.21
P25	-283276.96	P25	197149.74
P75	470176.82	P75	277242.59
P99	877680.95	P99	311391.54
Min	-1915353.33	Min	-2718988.81
Max	903268.60	Max	315293.57

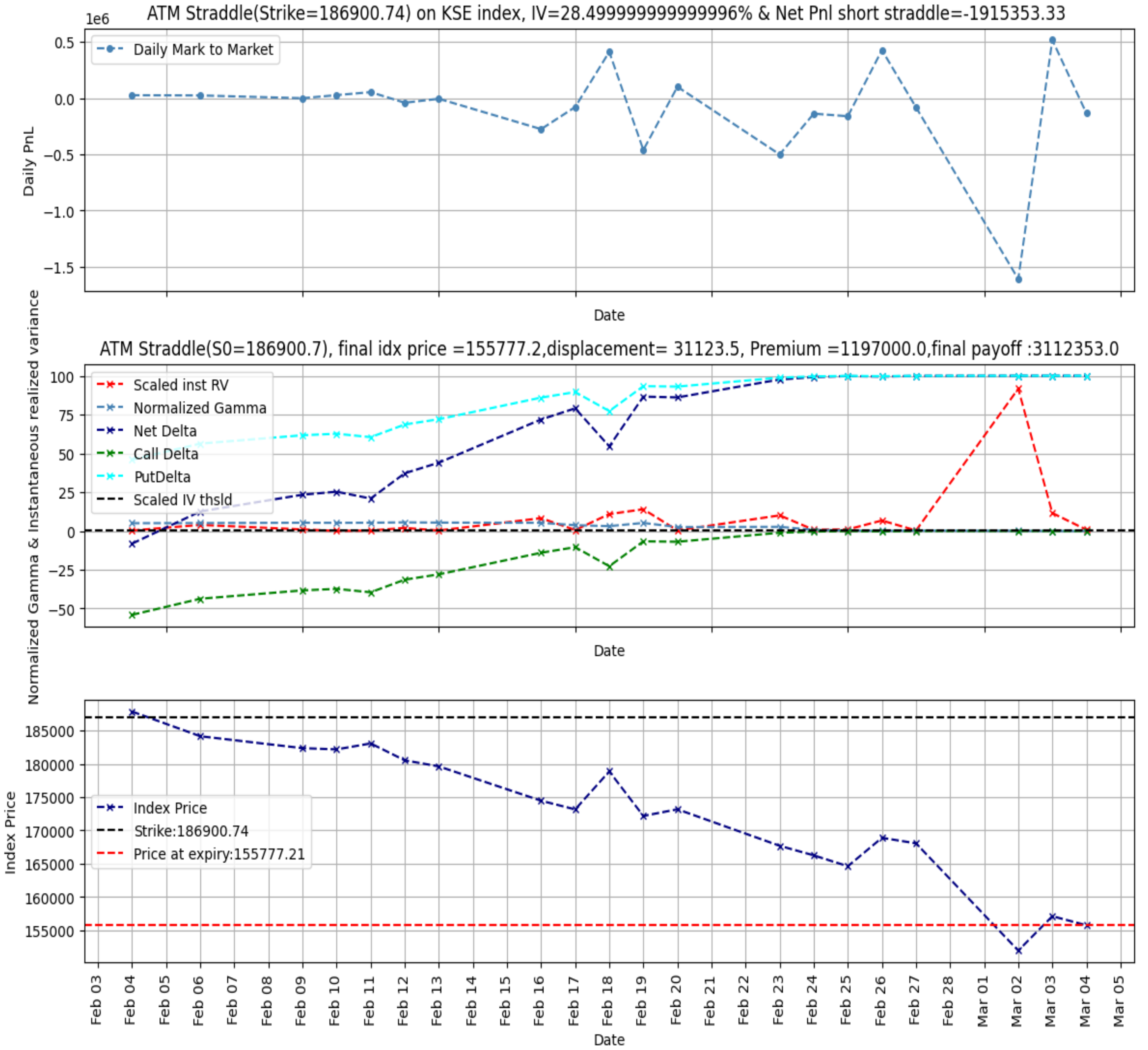




Hedge or Not To Hedge? Market Under Negative Drift

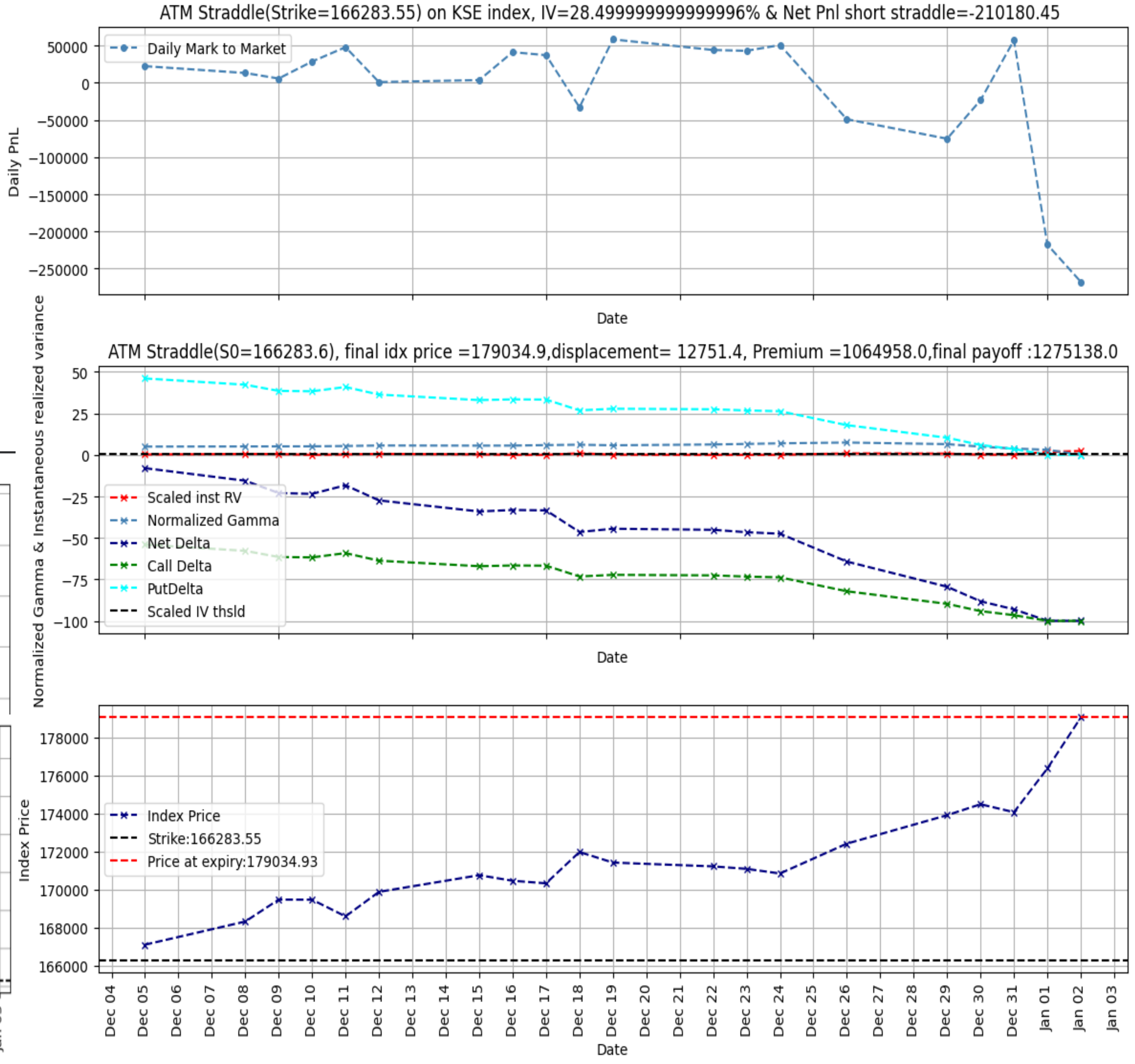
- Short ATM unhedged straddle incurs a worst case Rs. 2Million loss in the month of Feb 2026 when market has a consistent negative drift. Notice that short straddle strategy's net delta is on the rally so: *<u>a delta neutral strategy would require shorting more and more shares as index price plummets and buying them back at essentially the lowest price, consequently minimizing loss by (1/5)th shown below:</u>*

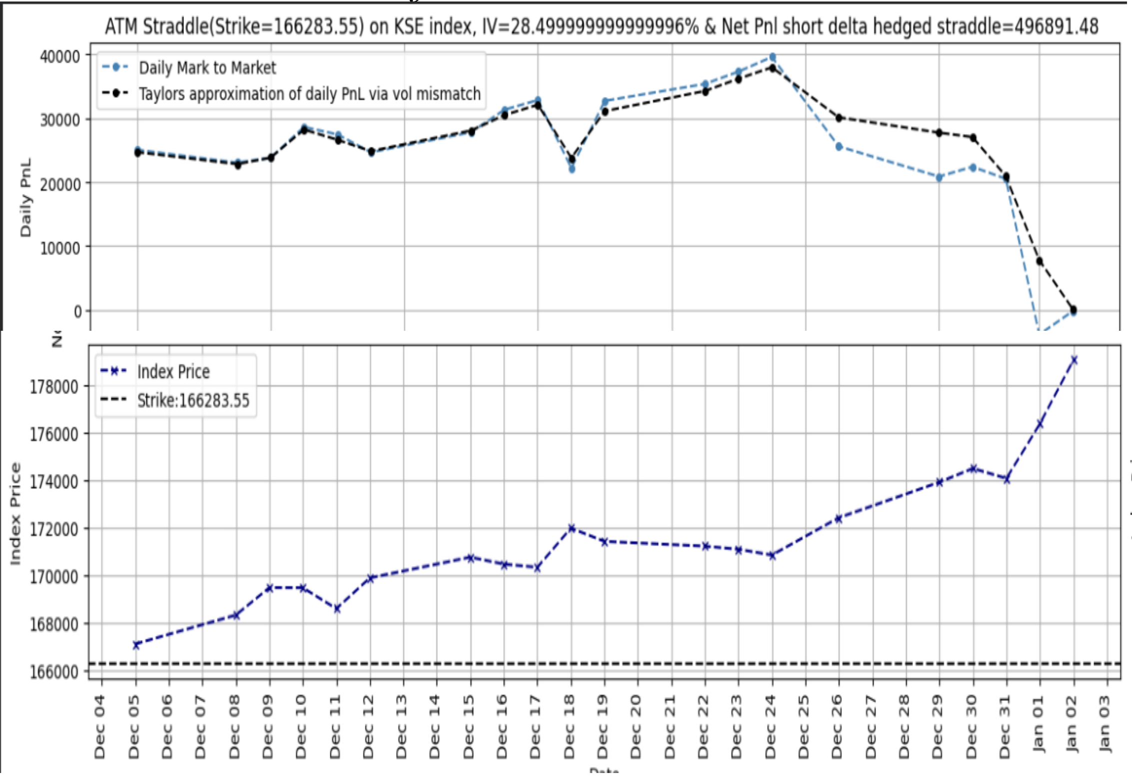




Hedge or Not To Hedge? Market Under Positive Drift

- Short ATM unhedged straddle incurs a loss Rs. 0.2Million loss in the month of Dec 2025 when market has a consistent positive drift. Notice that short straddle strategy’s net delta is decreasing: *a delta neutral strategy would require buying more and more shares as index price raises and selling them at highest price, consequently turning the loss into Rs. 0.5Million Profit shown below:*



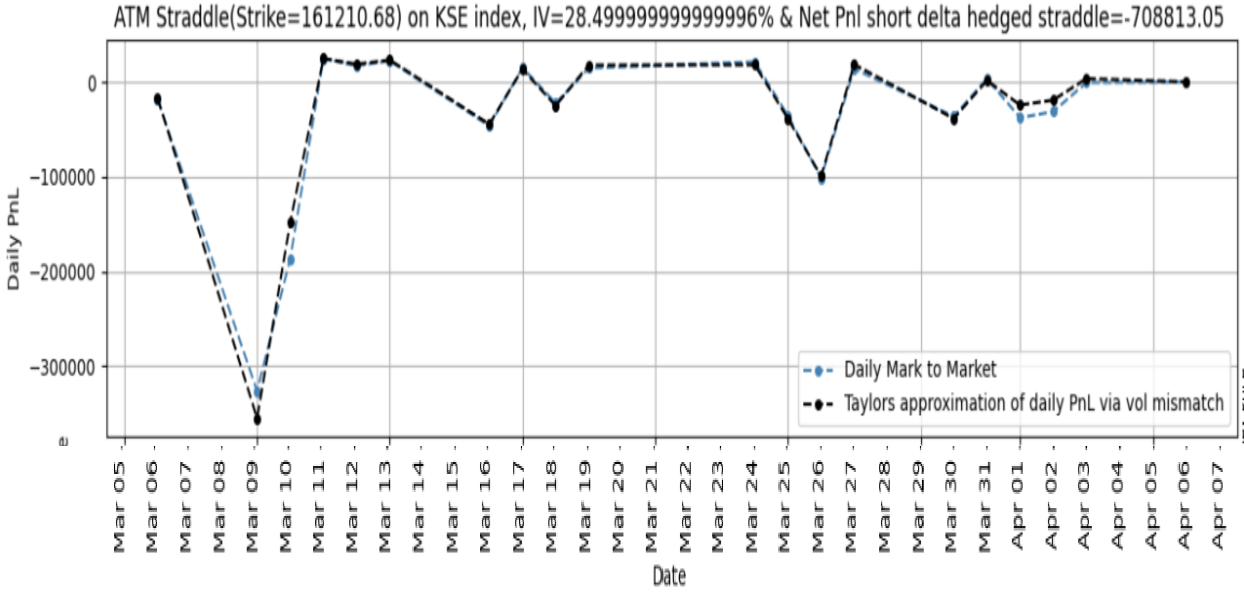


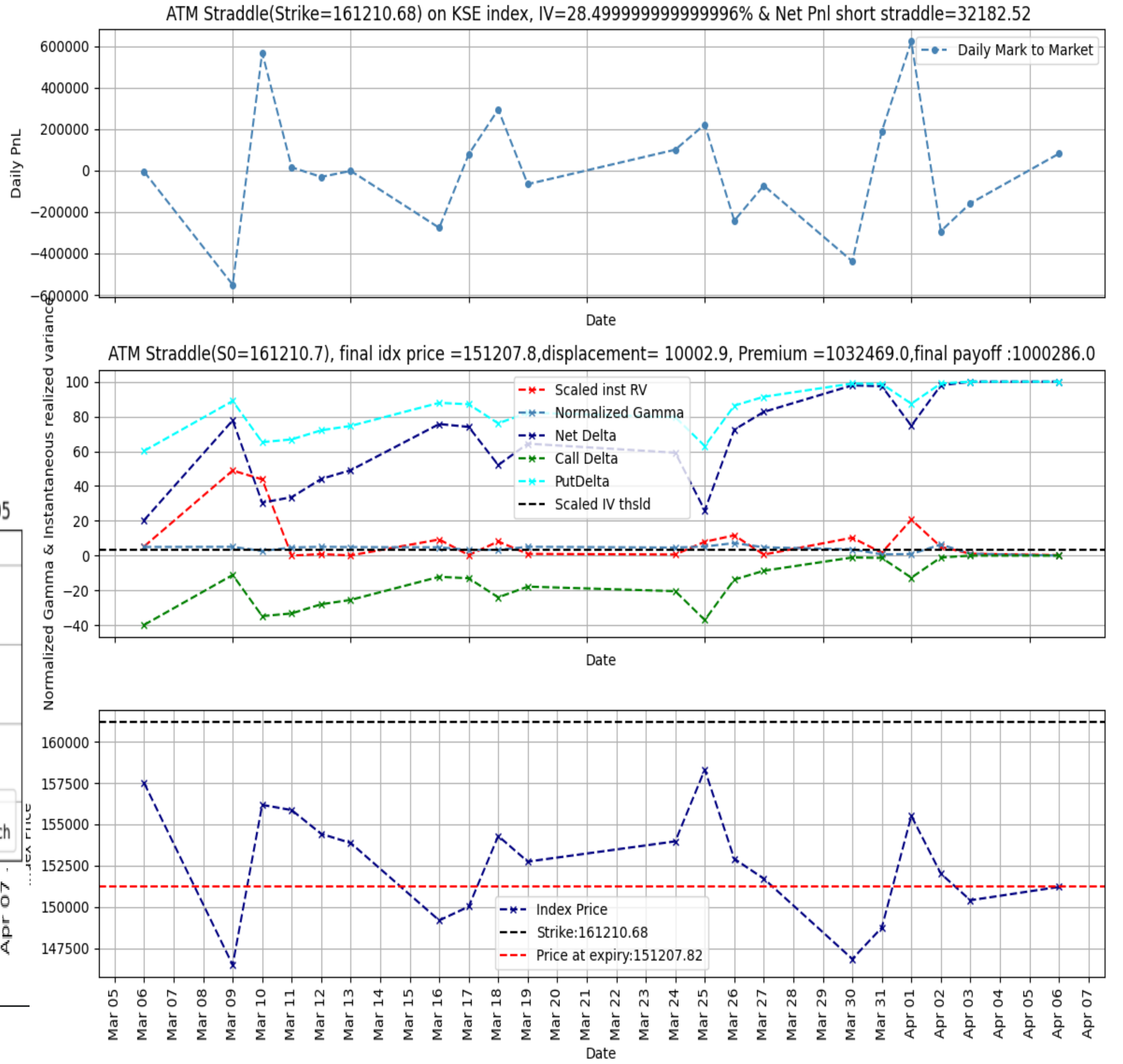
Hedge or Not To Hedge? Choppy Driftless Market

- Short ATM unhedged straddle shows a profit of Rs. 32000 in during the month of March 2026; index price is oscillating around the strike hence delta hedging only leaves exposure to losses due to volatility mismatches

$$\left(\frac{S_t - S_{t-1}}{S_{t-1}}\right)^2 > \sigma_{iv}^2\left(\frac{1}{252}\right) \qquad \& \qquad \Gamma \gg 0$$

- <u>*Delta Hedging in highly volatile driftless market turned a Rs.30000 profit into a Rs 0.7Million loss.*</u>

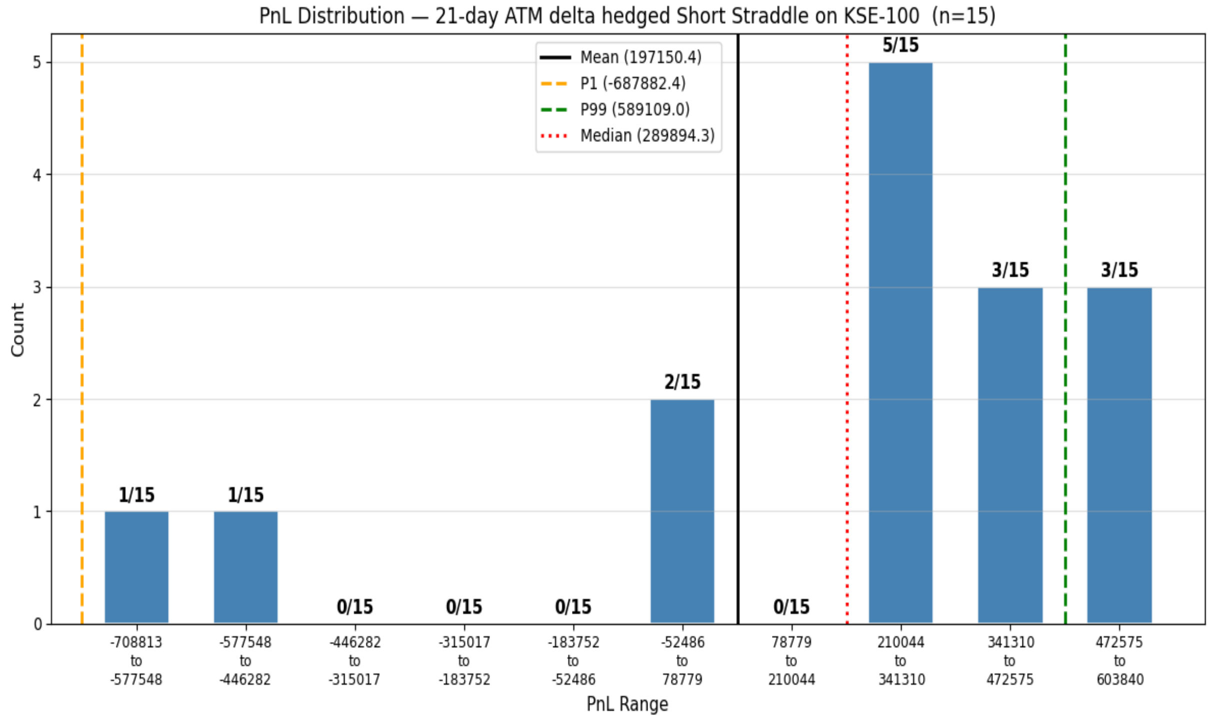




Margin Adjustments For Delta Hedged Positions: Straddle

- As seen in preceding examples, one can not make an absolute statement about lower loss probability for a delta hedge short position.
- But on average: delta hedged short straddle reduces the max loss by approximately 2.7

Margin for naked short straddle=2.7 × Margin for hedged short straddle

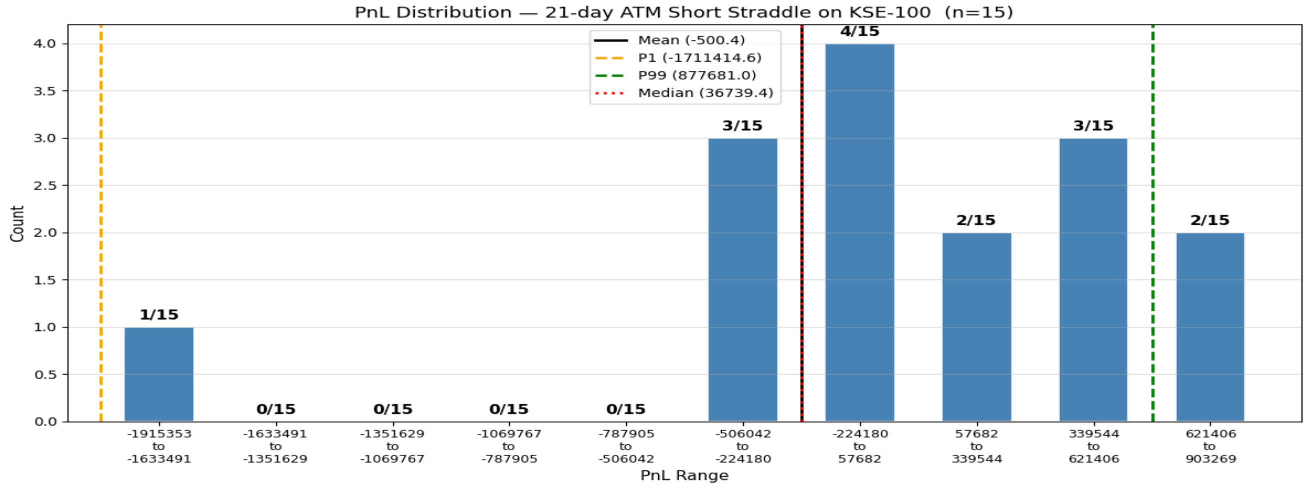


PnL Statistics — 21-day ATM Short Straddle

Statistic	Value
Samples	15.00
Mean	-500.36
Std Dev	655598.66
Median	36739.37
Skewness	-1.34
Ex Kurtosis	2.32
P1	-1711414.55
P25	-283276.96
P75	470176.82
P99	877680.95
Min	-1915353.33
Max	903268.60

PnL Statistics — 21-day ATM delta hedged Short Straddle

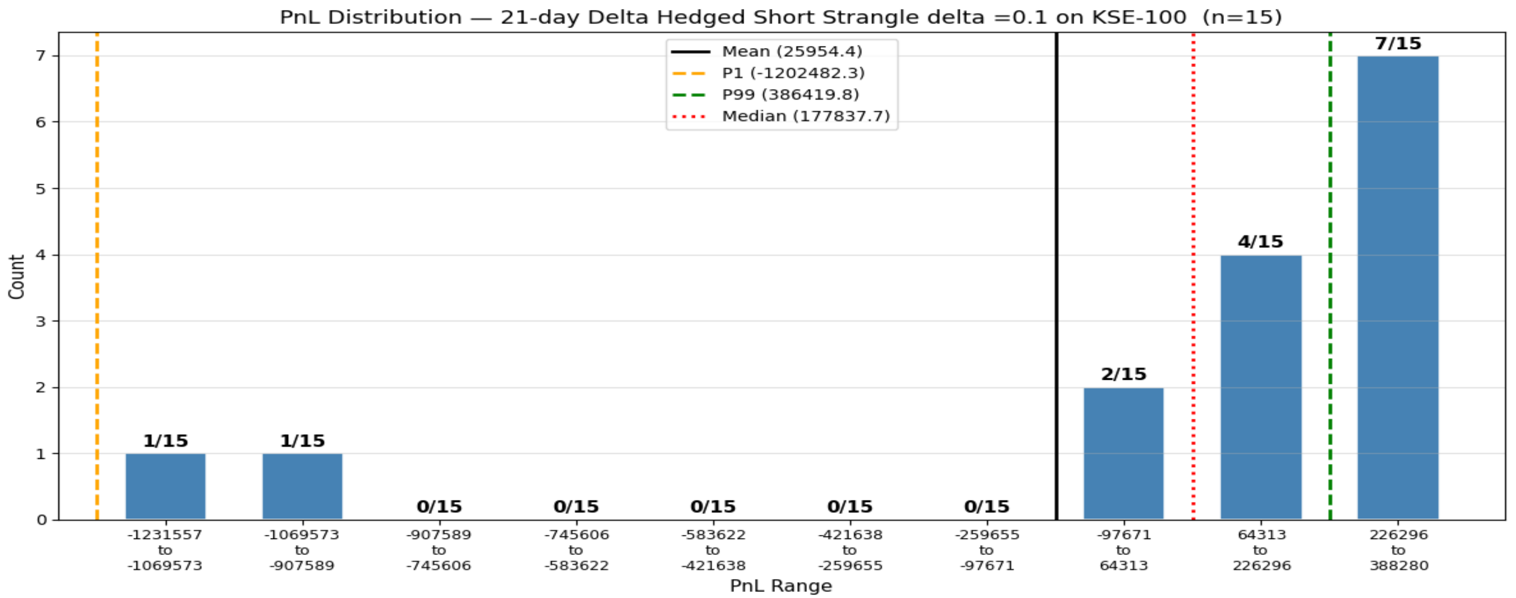
Statistic	Value
Samples	15.00
Mean	197150.43
Std Dev	370291.79
Median	289894.25
Skewness	-1.36
Ex Kurtosis	0.76
P1	-687882.37
P25	99520.98
P75	448576.38
P99	589109.05
Min	-708813.05
Max	603840.17

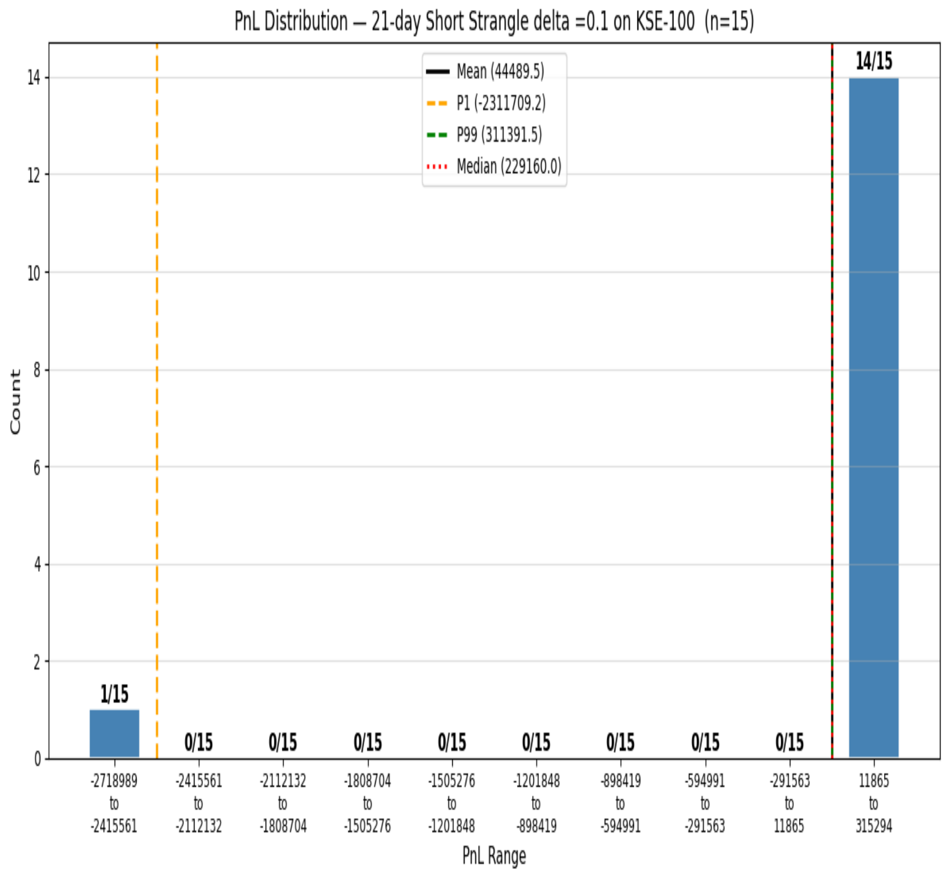


Margin Adjustments For Delta Hedged Positions: Strangle

- On average: delta hedged short strangle reduces the max loss by approximately 2.2

Margin for naked short strangle=2.2 × Margin for hedged short strangle





PnL Statistics — 21-day Init 0.1 delta Short Strangle

Statistic	Value
Samples	15.00
Mean	44489.54
Std Dev	739680.48
Median	229160.00
Skewness	-3.46
Kurtosis (exc)	9.99
P1	-2311709.21
P25	197149.74
P75	277242.59
P99	311391.54
Min	-2718988.81
Max	315293.57

PnL Statistics — 21-day Init 0.1 delta Short Strangle(Hedged)

Statistic	Value
Samples	15.00
Mean	25954.42
Std Dev	470743.28
Median	177837.69
Skewness	-1.90
Kurtosis (exc)	2.12
P1	-1202482.29
P25	76534.61
P75	258500.71
P99	386419.79
Min	-1231556.52
Max	388279.94

Bleed(Long Vol) or Blowup(Short Vol)? Why Do We Prefer Asymmetric Payoffs ?

- **Skewness Risk:** In forecasting models utilized in the financial field is the risk that results when observations are not spread symmetrically around an average value, but instead have a skewed distribution. As a result, the mean and the median can be different. Skewness risk can arise in any quantitative model that assumes a symmetric distribution (such as the normal distribution) but is applied to skewed data.

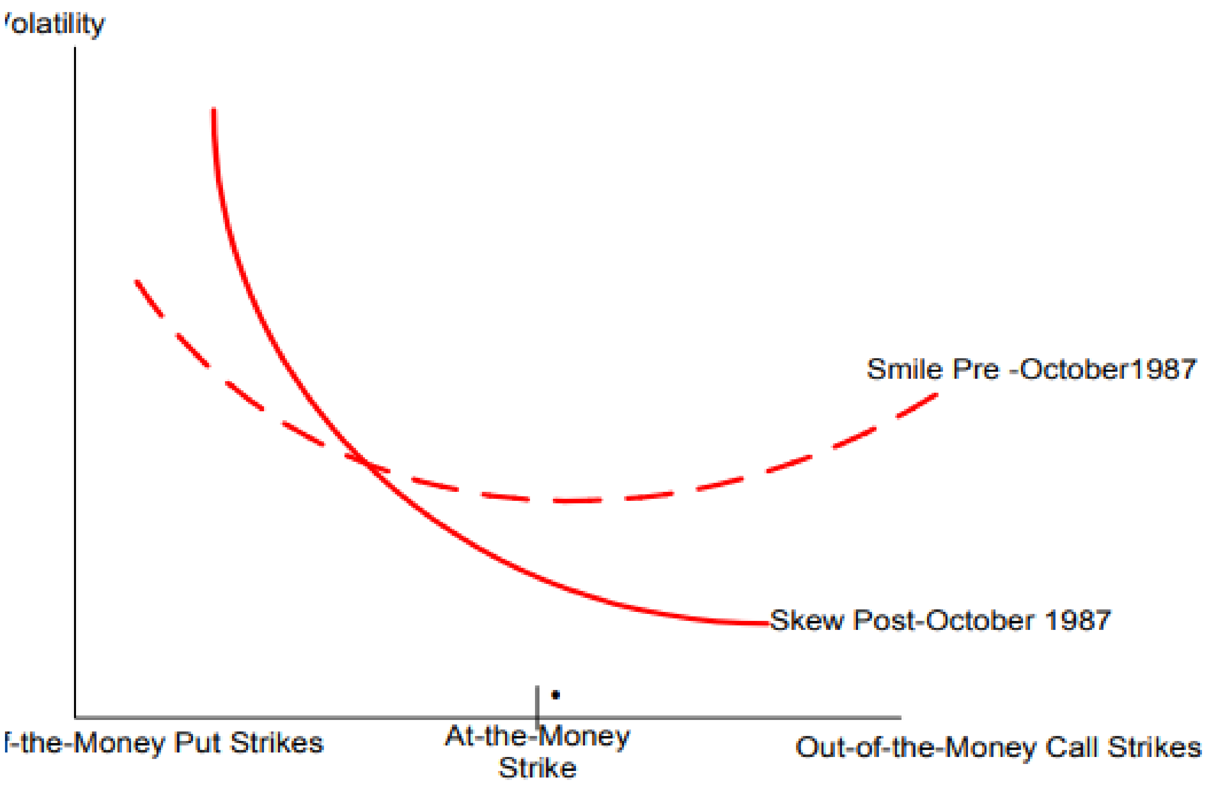
- **Kurtosis Risk:** Kurtosis risk is commonly referred to as "fat tail" risk. The "fat tail" metaphor explicitly describes the situation of having more observations at either extreme than the tails of the normal distribution would suggest; therefore, the tails are "fatter". Ignoring kurtosis risk will cause any model to understate the risk of variables with high kurtosis. For instance, Long-Term Capital Management, a hedge fund cofounded by Myron Scholes, ignored kurtosis risk to its detriment. After four successful years, this hedge fund had to be bailed out by major investment banks in the late 1990s because it understated the kurtosis of many financial securities underlying the fund's own trading positions.[1][2]

- **The smooth ride effect(Short Vol)** The process generates steady, comfortable returns with low day-to-day volatility — then concentrates all the variance in a single brief catastrophic period. This makes the strategy *feel* safe and low-risk.

The Two Strategies	Nero (Bleed)	Carlos (Blowup)
99 weeks	Loses $1,000/week	Gains $1,000/week
Week 100	Wins $99,000	Loses $99,000
Math expectation	Same	Same
Skewness	**Positive**	**Negative**
Quality of life	**Miserable most of the time**	**Happy most of the time**

- Same expected value. Carlos has a better *life experience* despite identical math — this is Taleb's central paradox why there always be option seller

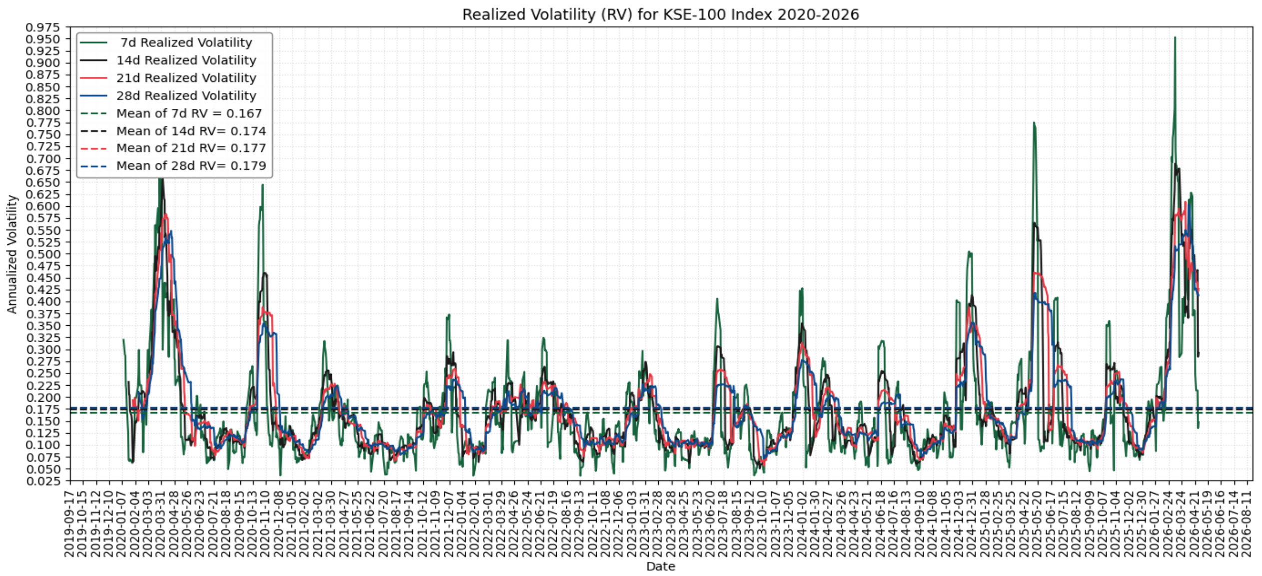
Financial Mathematics

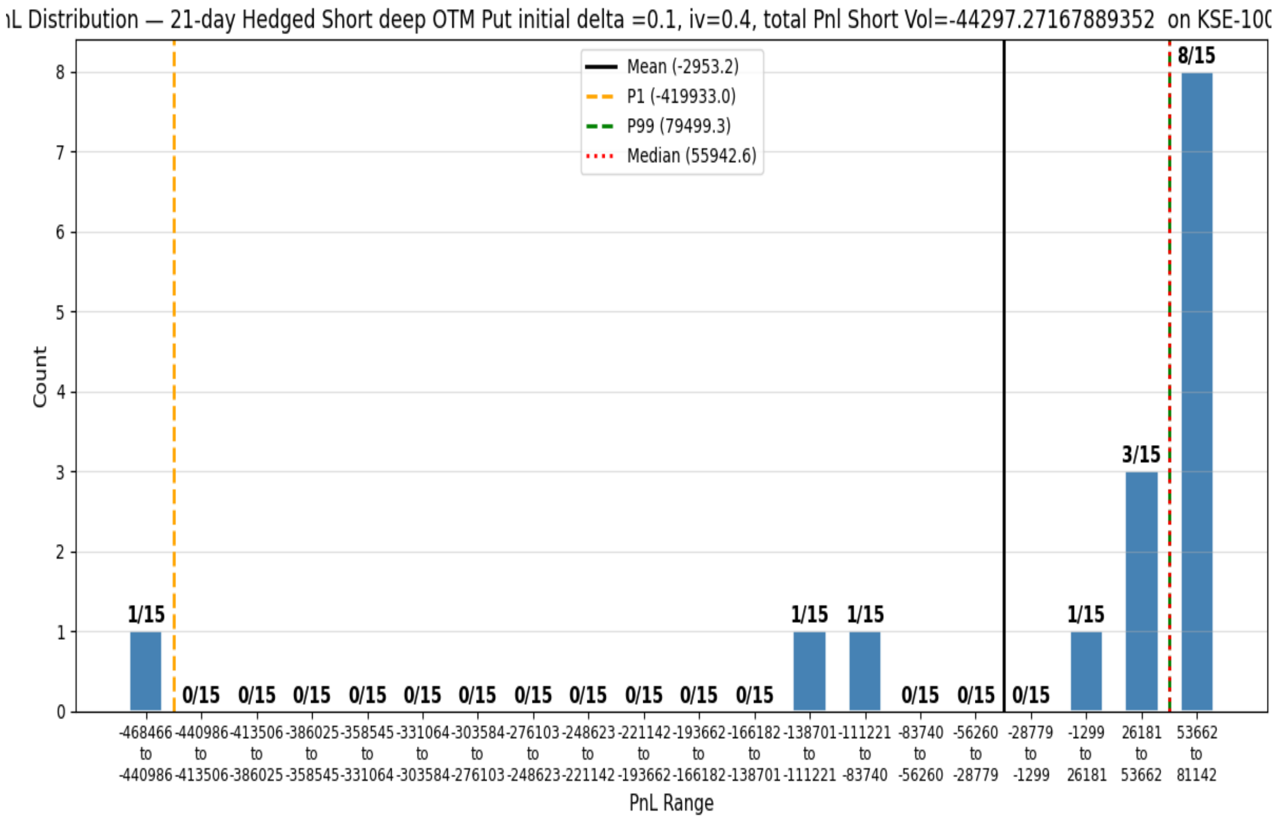


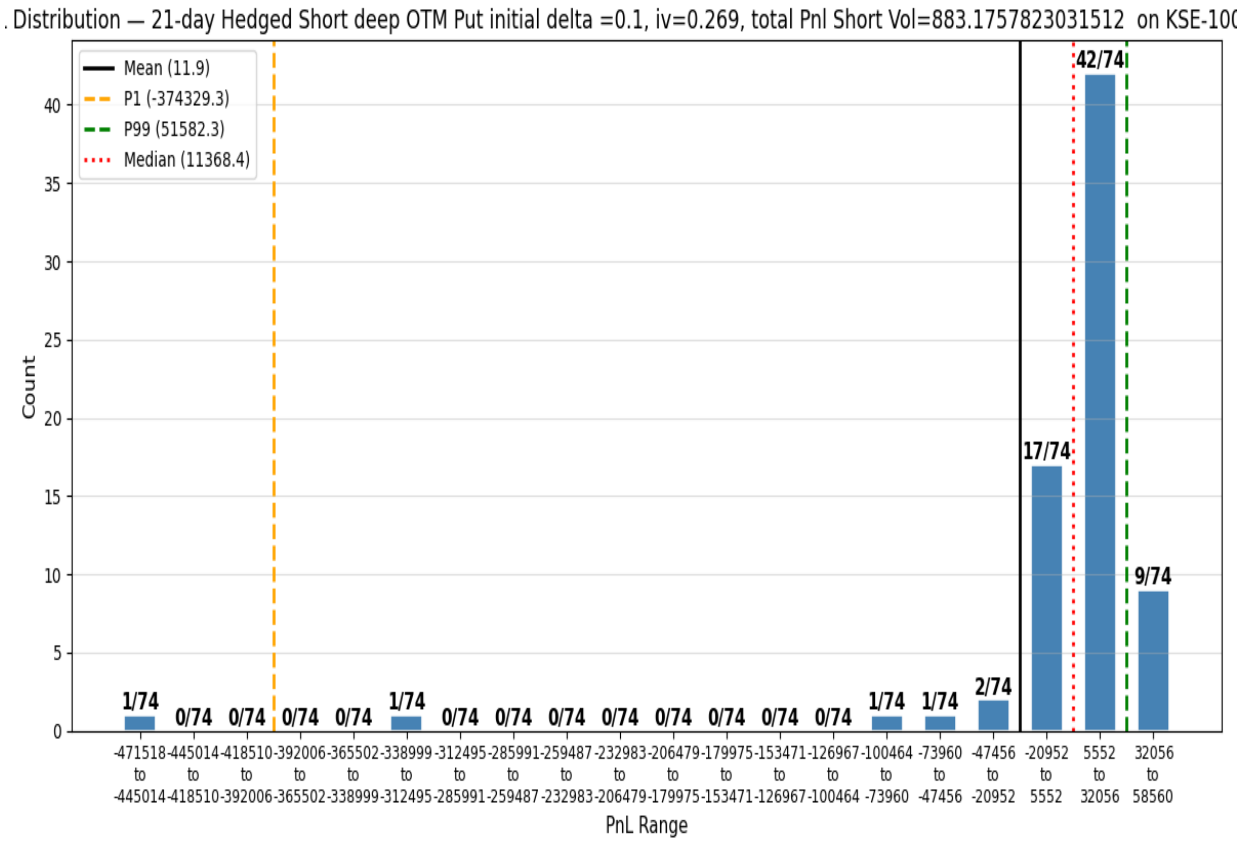
S&P 500 options implied volatility before and after 1987 crash

Index Volatility On The Rise

- Realized volatility KSE-100 index is on the rise, and so is the break-even implied volatility: short delta-hedged 21 day put positions from 2020-2026 have a 10-delta break-even volatility of around 26.9%, whereas from 2025-2026 it is more than 42%.
- How to leverage on this fact: Long volatility pays off when index plummets. So how to introduced volatility as an asset class?







CBOE VIX

- VIX is expected future realized volatility of S&P500 index over next 30 days. It is calculated from the prices of OTM options on S&P500 expiring around 30 days.
- Index returns and implied volatility are negatively correlated: right after a crash people realize consequences of not buying insurances in the form of Deep OTM Put. High demand raises the price of Deep OTM Put's and consequently VIX level. Negative correlation/ leverage effect in U.S Equity Market: when S&P500(SPX) is at its' lowest on Mar 26, VIX(CBOE implied volatility index is at the highest)

Risk Neutral(RN) Expected Future Realized Volatility

- Under risk-neutral measure $\mathbb{Q}$: $dS_t = rS_t dt + \sigma_t S_t dW_t$

$$\ln\left(\frac{S_T}{S_0}\right) = rT + \int_0^T \sigma_t dW_t - \frac{1}{2}\int_0^T \sigma_t^2 dt \quad \& \quad \mathbb{E}_{\mathbb{Q}}[\ln S_T] = \ln\left(S_0 e^{rT}\right) - 0.5\mathbb{E}_{\mathbb{Q}}\left[\int_0^T \sigma_t^2 dt\right] \ \& \ \text{at } t = 0\text{: } \mathbb{E}_{\mathbb{Q}}[S_T] = e^{rT}S_0 =: F$$

- **Theorem**: Let $f(x) \in C^2(a,b)$, given $x_0 \in (a,b)$ for all $x \in (a,b)$

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \int_a^{x_0} (y - x)^+ f''(y)dy + \int_{x_0}^{b} (x - y)^+ f''(y)dy$$

- Any European style derivative can be priced via call and put options:

$$f(S_T) = f(K_0) + (S_T - K_0)f'(K_0) + \int_0^{K_0}(K - S_T)^+ f''(K)dK + \int_{K_0}^{\infty}(S_T - K)^+ f''(K)dK$$
$$= f(K_0) + (S_T - K_0)f'(K_0) + \int_0^{K_0} P_K(S_T)f''(K)dK + \int_{K_0}^{\infty} C_K(S_T)f''(K)dK$$

- Let f be the natural log $\Rightarrow \ln S_T = \ln K_0 + \left(\frac{S_T}{K_0} - 1\right) - \int_0^{K_0} \frac{P_K(S_T)}{K^2} dK - \int_{K_0}^{\infty} \frac{C_K(S_T)}{K^2} dK$

- At inception t=0, let Put(K) and Call(K) be the put and call prices with strike K expiring at future time T:

$$\mathbb{E}_{\mathbb{Q}}[\ln S_T] = \ln K_0 + \left(\frac{F}{K_0} - 1\right) - \int_0^{K_0} \frac{e^{rT}\mathrm{Put}(K)}{K^2} dK - \int_{K_0}^{\infty} \frac{e^{rT}\mathrm{Call}(K)}{K^2} dK$$

- **RN expected future realized volatility**

$$\mathbb{E}_{\mathbb{Q}}\left[\int_0^T \sigma_t^2 dt\right] = 2\ln\left(\frac{F}{K_0}\right) + 2\left(1 - \frac{F}{K_0}\right) + 2\int_0^{K_0} \frac{e^{rT}\mathrm{Put}(K)}{K^2} dK + 2\int_{K_0}^{\infty} \frac{e^{rT}\mathrm{Call}(K)}{K^2} dK$$

Volatility Index From Options On Index

- **RN expected future realized volatility**

$$\widehat{\sigma_T^2} = \frac{2}{T}\left(\ln\left(\frac{F}{K_0}\right) + 1 - \frac{F}{K_0}\right) + \frac{2}{T}\int_0^{K_0} \frac{e^{rT}\mathrm{Put}(K)}{K^2}dK + \frac{2}{T}\int_{K_0}^{\infty} \frac{e^{rT}\mathrm{Call}(K)}{K^2}dK \approx -\frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 + \frac{2}{T}\int_0^{K_0} \frac{e^{rT}\mathrm{Put}(K)}{K^2}dK + \frac{2}{T}\int_{K_0}^{\infty} \frac{e^{rT}\mathrm{Call}(K)}{K^2}dK$$

- **CBOE $\mathbf{K_0}$** $:= \max_K\{K \leq F\} = \max_K\{C_K \geq P_K\}$, so CBOE finds K_0 operationally by scanning the option chain and fi last strike where the call is still more expensive than put; the F is recovered via put-call parity: $F = e^{rT}(C_{K_0} -$ K_0.

- **If an immediate 21 day tenor available: VIX=100$\sqrt{\widehat{\sigma^2_{T=21-\text{day tenor}}}}$**

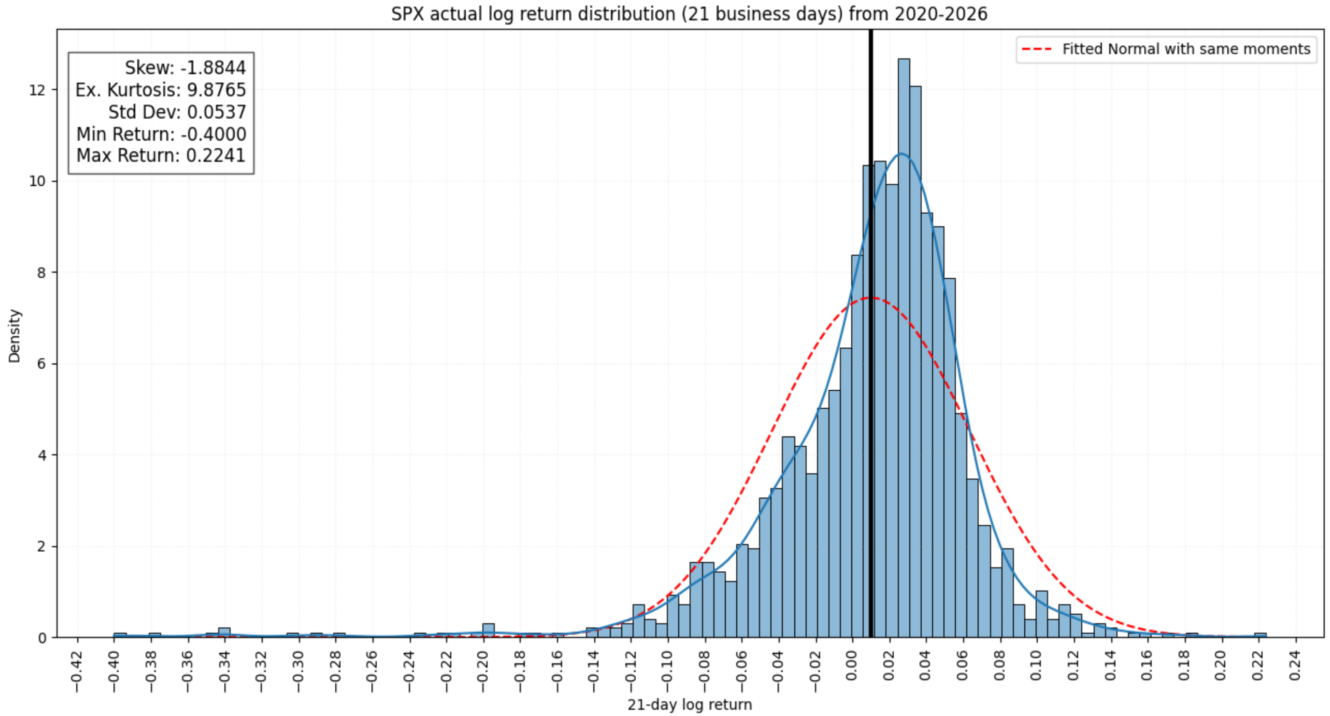
$$\widehat{\sigma_T^2} = \frac{2}{T}\sum_{K_i<K_0}\left(\frac{e^{rT}\,P_{K_i}}{K_i^2}\right)\left(\frac{K_{i+1}-K_{i-1}}{2}\right) + \frac{2}{T}\sum_{K_i>K_0}\left(\frac{e^{rT}\,C_{K_i}}{K_i^2}\right)\left(\frac{K_{i+1}-K_{i-1}}{2}\right) + \frac{1}{T}\left(\frac{e^{rT}\,P_{K_0} + e^{rT}\,C_{K_0}}{K_i^2}\right)\left(\frac{K_1-K_{-1}}{2}\right) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2$$

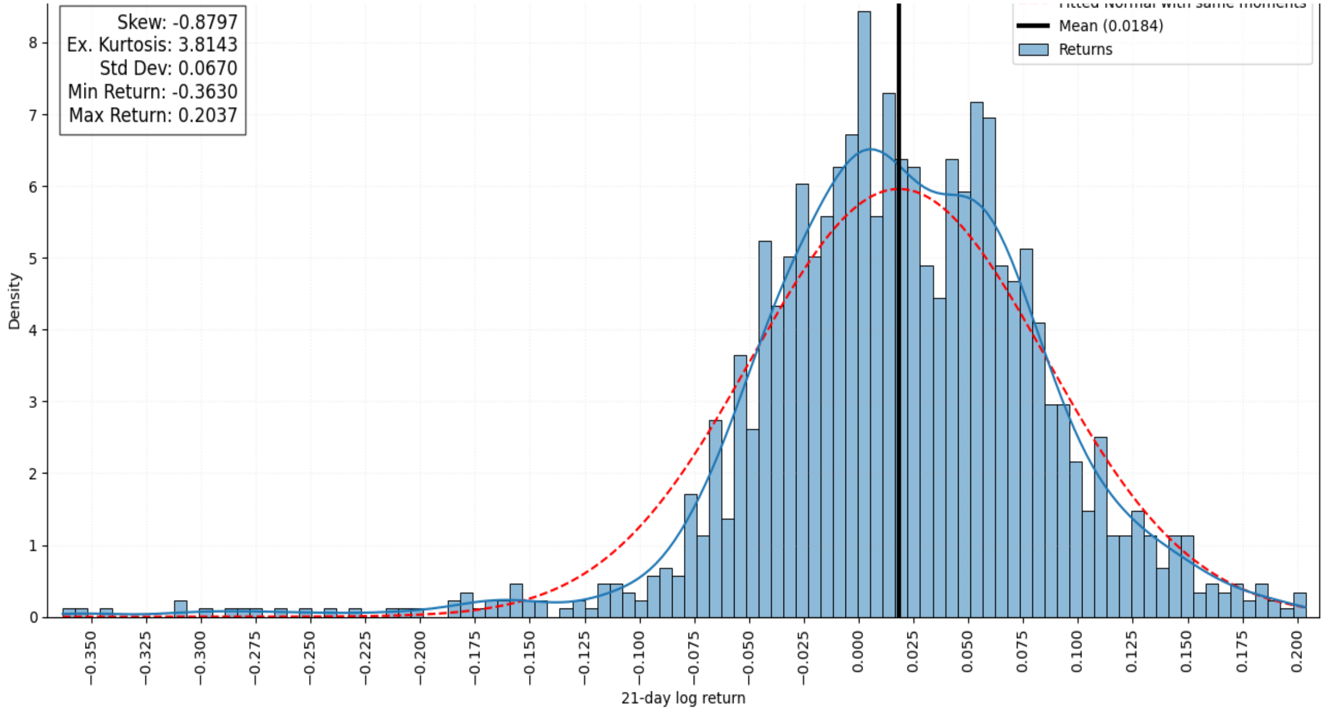
- **Otherwise:** interpolation via front and back month total variances see

Skew Premium In Returns

- **Realized Return Skew:** Index return follows <u>Taleb's distribution</u> i.e., an <u>investment returns</u> profile in which there is a high <u>probability</u> of a small gain, and a small probability of a <u>very large loss, which may more than outweighs the gains</u>; evident from negative skews ranging between -0.31 to -1.15 for maturities ranging from 1 business day to 28 business days.
- **Skew Premium:** As a compensation for providing crash protection, option pricing be as such that magnitude of implied skew deduced from index option prices satisfy:
- <u>Skew(Realized Index Log Returns)> Skew(Implied Index Log Returns)</u>

Tenor(Business Days)	Skew in Realized Index Log Returns: (KSE-100, SPX)
1	(-0.31,-0.64)
7	(-1.07,-1.37)
14	(-1.15,-1.53)
21	(-0.88,-1.88)
28	(-0.71,-1.73)
35	-0.6
49	-0.37
53	(-0.31,-0.99)
60	-0.08
80	(0.16,-0.35)
90	0.14
115	0.09
140	0.08
160	(0.02,-0.67)





Blowup Risk /Extra Margins Check Via Implied Returns Skew

- Let R_T be log return $R_T = \ln\left(\frac{S_T}{S_0}\right)$,then we can write $S_T = S_0 e^{R_T}$ which means that:

$$\mathbb{P}_{\mathbb{Q},S_0}(R_T \le x) = \mathbb{P}_{\mathbb{Q},S_0}\left(\ln\left(\frac{S_T}{S_0}\right) \le x\right) = \mathbb{P}_{\mathbb{Q},S_0}(S_T \le S_0 e^x) \approx \int_0^{S_0 e^x} \frac{\partial^2 C}{\partial K^2}(T,K)\,dK$$

$$\frac{d}{dx}\ \mathbb{P}_{\mathbb{Q},S_0}(R_T \le x) \approx \frac{d}{dx}\int_0^{S_0 e^x} \frac{\partial^2 C}{\partial K^2}(T,K)\,dK = \frac{d}{dx}\left(\frac{\partial C}{\partial K}(T, K = S_0 e^x) - \frac{\partial C}{\partial K}(T, K = 0)\right)$$

- Risk neutral returns over T days have density

$$\mathbb{P}_{\mathbb{Q},S_0}(R_T = x) = \frac{d}{dx}\frac{\partial C}{\partial K}(T, K = S_0 e^x) = \frac{1}{\sqrt{2\pi\sigma_{IV}^2 T}}\, e^{-\frac{\left(x+\frac{\sigma_{IV}^2}{2}T\right)^2}{2\sigma_{IV}^2 T}}$$

- If options are listed w.r.t multiple strikes we can use splines to deduce implied probability density function, of returns over next T days, under risk neutral measure and raise margins for option seller contradicting <u>Skew(Realized Index Log Returns)> Skew(Implied Index Log Returns)</u>